

4

Nature of Waves

In this chapter, we will learn about

- how waves are produced and how they transmit energy
- how particle vibrations give rise to wave motion
- the various terms and graphs used to describe particle vibrations and wave motion

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4.1 Introduction to waves

Flipped classroom
Watch this and answer the questions.



Fig 4.1a Seeing and hearing involve waves.

Cross-link
We will discuss light waves in Ch 6 and sound waves in Ch 7.

You have been sensing the world through different kinds of **waves** since you were born. The light entering your eyes is a kind of wave. The sound entering your ears is another kind of wave. Waves are so important to us.

DSE goal
Interpret wave in terms of oscillation.

1 Producing waves

When you tap the surface of water, the water surface vibrates and a water wave is formed (Fig 4.1b). When you move a rope up and down, the rope vibrates and a wave is formed on it. In fact, all waves can be described as **vibrations** or **oscillations**.



Fig 4.1b A water wave and a wave on a rope.

Cross-link
We will discuss the waves that do not travel from one place to another in Ch 5.
We will discuss the waves that do not need a medium to travel in Ch 6.

The waves shown in Figure 4.1b can travel—they travel away from the vibrating sources. Such waves are called **travelling waves**. Moreover, both waves need a medium to travel (water and a rope). They are *mechanical waves*.

wave 波　vibration/oscillation 振動　travelling wave 行波　mechanical wave 機械波

DSE goal

Realize waves as transmitting energy without transferring matter.

DSE exam

21(1A)Q10

2 Energy transmission in waves

In Figure 4.1c, when we push a box towards a stationary ball, the box moves forwards to hit the ball and makes the ball move. Energy is transmitted by transferring matter (box) to the ball.

In Figure 4.1d, when we flick a spring to produce a wave, the spring does not move forwards but it can still make the stationary ball move. This time, energy is transmitted to the ball without transferring any matter. The energy is transferred along the direction of travel (*propagation*) of the wave.

Video

→ This video demonstrates how energy is transmitted with or without transferring matter.

✎ Ss may notice that waves flush rubbish to the shore on a beach. Ts may explain that water movement has 2 components: water waves and water current. It is the water current that brings the rubbish to the shore.

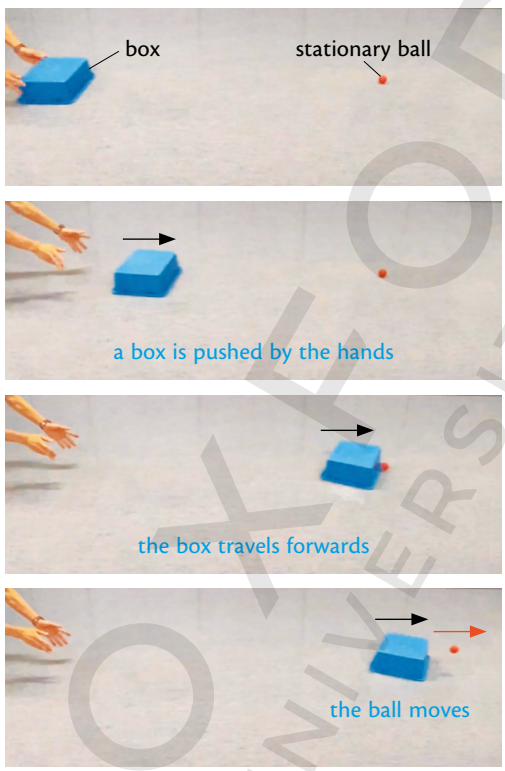


Fig 4.1c *Transmission of energy by a box.*

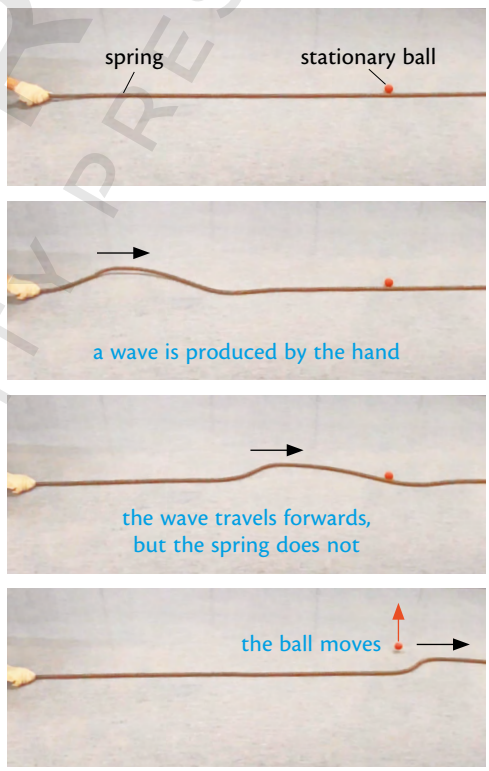


Fig 4.1d *Transmission of energy by a wave.*

Waves transmit energy without transferring matter.

Know more

Disasters caused by waves

When an earthquake occurs, a huge amount of energy is transmitted in the form of waves. The ground is shaken by the waves and may cause serious destruction.

An earthquake occurring beneath the sea displaces a large volume of water and a *tsunami* may be produced.



propagation 傳播　　tsunami 海嘯

DSE goal

Distinguish between transverse and longitudinal waves.

Describe wave motion in terms of waveform, crest, trough, compression and rarefaction.

Simulation Video

→ This simulation shows a travelling transverse wave made up of particles vibrating at right angles to the direction of travel of the wave. Ss can choose from 4 particles to a chain of particles to observe the progression of the transverse wave.

→ This video demonstrates the properties of transverse pulses and waves on a slinky spring.

3 Transverse and longitudinal waves

Based on the relative direction of vibration and the direction of travel, we can categorize waves into two types, namely **transverse waves** and **longitudinal waves**.

In Figure 4.1e, a slinky spring is stretched and one of its ends is fixed. If we flick its free end sideways repeatedly, we produce a wave which travels towards the fixed end along the spring.

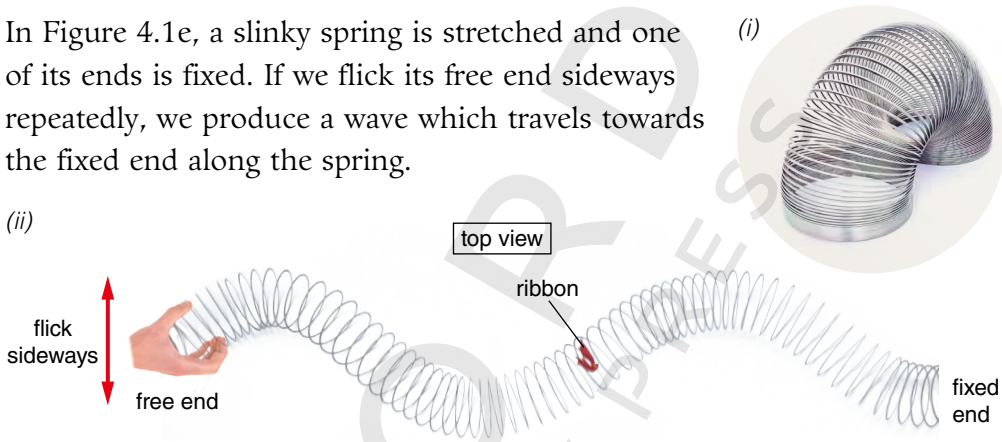


Fig 4.1e *(i) A slinky spring. (ii) Flicking a slinky spring sideways to produce a wave.*

The ribbon, which represents a particle on the spring, vibrates perpendicularly to the direction the wave travels (Fig 4.1f). Such kind of wave is called a transverse wave. Its **waveform** consists of a series of **crests** and **troughs** (Fig 4.1g).

Water waves and waves on a string are examples of transverse waves.

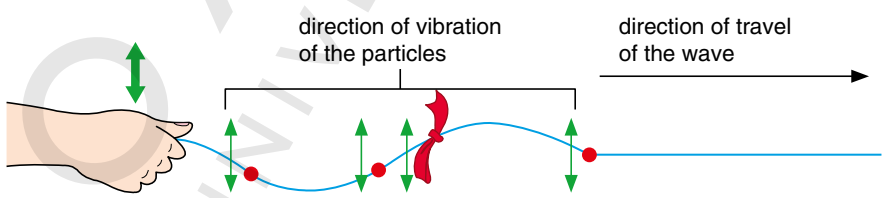


Fig 4.1f *On a transverse wave, particles vibrate perpendicularly to the direction the wave travels.*

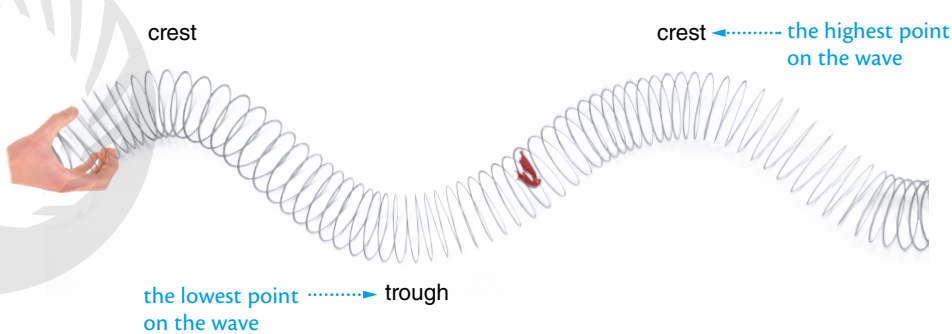


Fig 4.1g *Crests and troughs on a transverse wave.*

Note that when a wave travels forwards, a particle on the wave does not travel forwards with the wave. It only vibrates about a fixed position.

transverse wave 橫波　longitudinal wave 縱波　waveform 波形　crest 波峯　trough 波谷

Simulation Video

→ This simulation shows a travelling longitudinal wave made up of particles vibrating along the direction of travel of the wave.

→ This video demonstrates the properties of longitudinal pulses and waves on a slinky spring.

We can also produce a wave by pushing and pulling the free end of a stretched slinky spring repeatedly (Fig 4.1h).

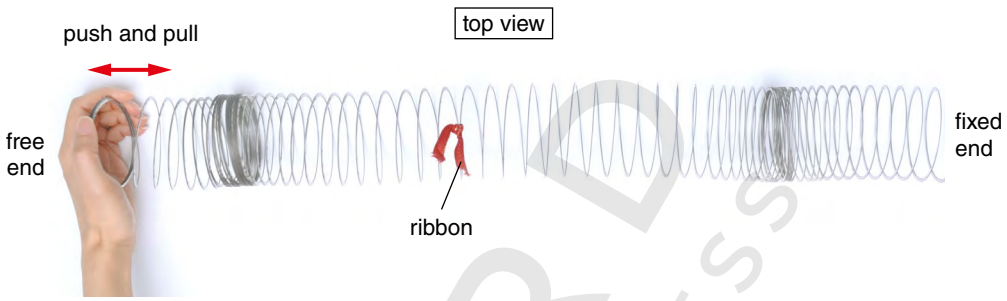


Fig 4.1h *Pushing and pulling a slinky spring to produce a wave.*

The ribbon vibrates along the direction the wave travels (Fig 4.1i). Such kind of wave is called a longitudinal wave. Its waveform consists of a series of **compressions** and **rarefactions** (Fig 4.1j).

Sound waves are examples of longitudinal waves.

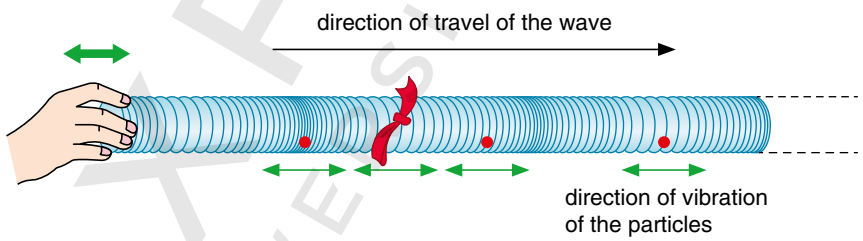


Fig 4.1i *On a longitudinal wave, particles vibrate along the direction the wave travels.*

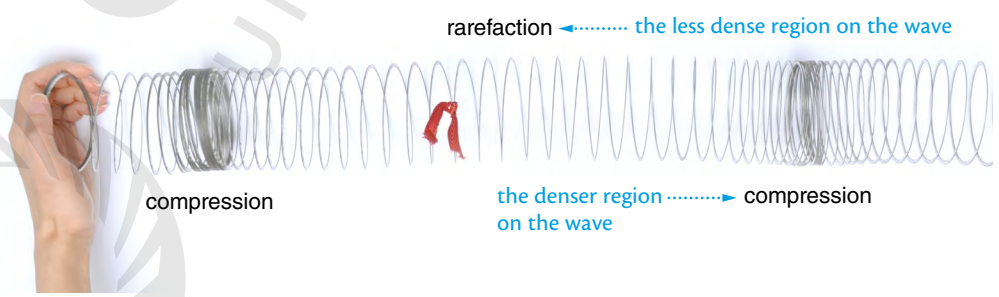


Fig 4.1j *Compressions and rarefactions on a longitudinal wave.*

On a transverse wave, the particle vibrations are perpendicular to the direction the wave travels.
On a longitudinal wave, the particle vibrations are parallel to the direction the wave travels.

compression 密部　　rarefaction 疏部

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Quick check 1

E-version in OUP Exercise Platform is also available.

1 What does a wave transmit?

A Energy

B Matter

C Energy and matter

2 In which kind of wave do the particles vibrate parallel to the direction the wave travels?

A Transverse wave

B Longitudinal wave

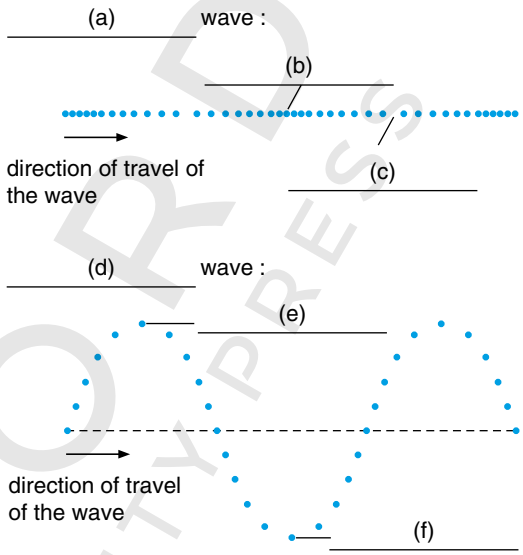
3 *True or false*:

(a) A particle on a longitudinal wave always remains stationary because the wave does not transfer matter. (T/F)

(b) You can hear a sound because energy is transmitted from the sound source to you in the form of waves. (T/F)

(c) All mechanical waves can be described as vibrations. (T/F)

4 Two waves are shown below. Fill in the blanks with the following terms: transverse, longitudinal, crest, trough, rarefaction, compression.



Exercise 4.1

★**1** A cork is floating on water. It moves after a stone is dropped into the water near it (Fig a).

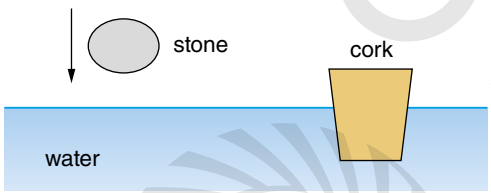


Fig a

Which of the following statements are correct?

(1) The stone produces a transverse wave on the water.

(2) The cork moves towards the right.

(3) Energy is transmitted from the stone to the cork.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

2 A hand holds one end of a slinky spring (Fig b).



Fig b

(a) How should the hand move to produce (i) a transverse wave and (ii) a longitudinal wave on the spring?

(b) How does a particle on the spring move in the above cases?

★**3** Explain the following terms: crest, trough, compression and rarefaction.

★**4** A toy duck is floating on water (Fig c). Describe what will happen to the toy duck when you tap the surface of water near it.



Fig c

4.2 Wave motion of transverse waves

1 Describing wave motion

DSE goal

Describe wave motion in terms of displacement, amplitude, period, frequency, wavelength and wave speed.

Apply $f = 1/T$ and $v = f\lambda$ to solve problems.

DSE exam

16(1A)Q16

Simulation

→ This simulation shows the meaning of amplitude, wavelength, period and frequency of a transverse wave. The size of each quantity can be changed to show how the wave is affected. It is available on OUP web in PC version (.exe) only.

In this part, we will learn about some useful terms to describe a transverse wave. Longitudinal waves will be discussed in Chapter 7.

a Amplitude (A)

Consider a stretched string. When there is no wave, the string is straight. This is the **equilibrium position** of the string. When a wave travels along the string, different positions have different displacements from their equilibrium positions (Fig 4.2a). The size of the maximum displacement is the **amplitude** A of the wave. Its unit is the metre (m).

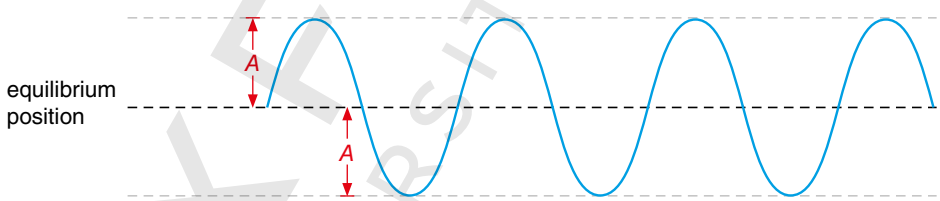


Fig 4.2a *Amplitude of a transverse wave.*

The amplitude of a transverse wave is the distance of the crest or the trough from the equilibrium position.

b Wavelength (λ)

The shortest distance that the wave repeats itself is the **wavelength** λ. Its unit is the metre (m).

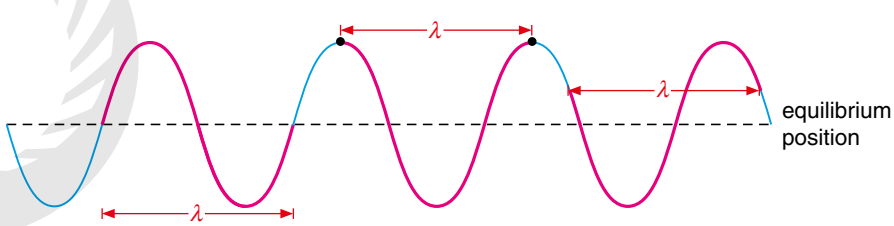


Fig 4.2b *Wavelength of a transverse wave.*

The wavelength of a transverse wave is equal to the distance between two successive crests (or troughs).

equilibrium position 平衡位置　　amplitude 振幅　　wavelength 波長

c Period (T)

Figure 4.2c shows a wave travelling to the right. From $t = 0$ to $t = T$, a complete cycle of the wave is produced, and the wave travels one λ. The time T is the **period** of the wave. Its unit is the second (s).

A complete cycle of a wave refers to the section of the wave in one wavelength. It is also called a complete wave.

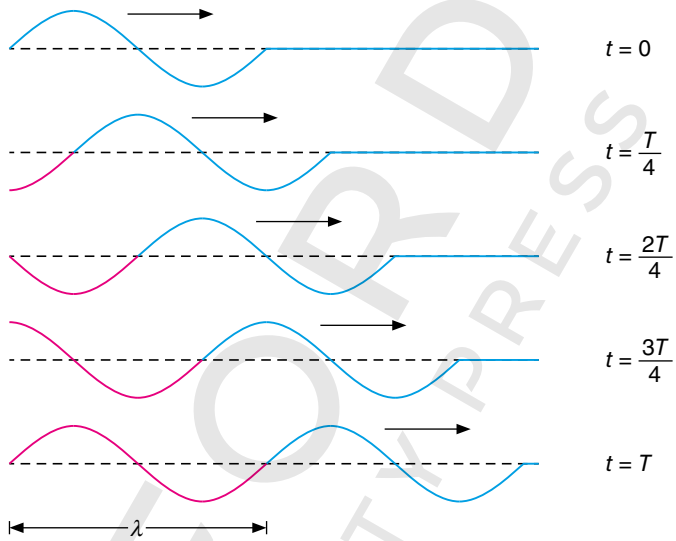


Fig 4.2c *The distance travelled by a wave in one period.*

The period of a wave is the time required to produce a complete cycle of the wave. It is also the time for the wave to travel one wavelength.

Note that **a wave travels a distance of $n\lambda$ in time nT.**

Know more

Heinrich Hertz (1857–1894)



The unit of frequency is named after the German scientist *Heinrich Hertz*, who discovered radio waves in 1886.

d Frequency (f)

Sometimes, instead of period, we prefer to describe a wave in terms of **frequency** f.

Tip
The frequency of a wave is the same as the frequency of the vibrating source.

The frequency of a wave is the number of complete cycles of the wave produced in one second.

Its unit is the **hertz** (**Hz**). If the frequency of a wave is 4 Hz, 4 complete cycles of the wave will be produced in one second. This means that the time required for producing one complete cycle of the wave, i.e. the period, is $\frac{1}{4}$ s. In fact,

1 Hz = 1 s^{-1}

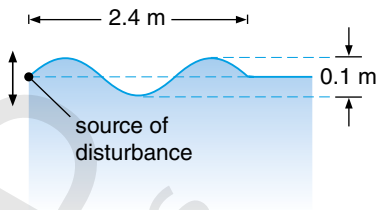
frequency (Hz) ⋯▸ $f = \frac{1}{T}$ ◂⋯ period (s)

This can also be expressed as $T = \frac{1}{f}$.

period 週期　frequency 頻率　hertz (Hz) 赫茲　Heinrich Hertz 赫茲

Example 1 Describing wave motion

A student produces waves on still water. He disturbs the water surface for 3 s and the waveform as shown is produced. Find the amplitude A, wavelength λ, period T and frequency f of the wave.



Solution

$$A = \frac{0.1}{2} = 0.05 \text{ m}$$

$$\lambda = 2.4 \times \frac{2}{3} = 1.6 \text{ m}$$

$$T = \frac{3}{1.5} = 2 \text{ s}$$

$$f = \frac{1}{T} = \frac{1}{2} = 0.5 \text{ Hz}$$

Tip

1.5 complete cycles are produced in 3 s.

Alternative method:

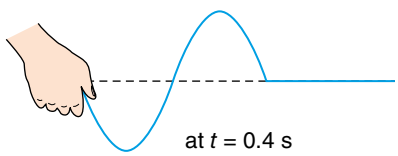
$$f = \frac{\text{no. of complete cycles}}{\text{time}}$$

$$= \frac{1.5}{3}$$

$$= 0.5 \text{ Hz}$$

Example 2 Sketching waveform

Susan flicks a rope up and down continuously to produce a wave with constant frequency and amplitude. She starts flicking at $t = 0$ and a complete cycle of the wave is produced at $t = 0.4$ s. Sketch the waveform of the wave at $t = 0.5$ s.



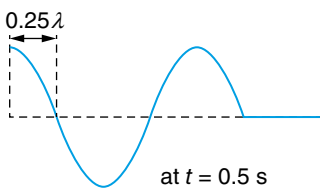
Solution

From $t = 0.4$ s to 0.5 s, the wave travels for 0.1 s.

As $T = 0.4$ s,

$0.1 \text{ s} = 0.25T$

$\therefore$ The wave moves to the right by 0.25λ.



→ Teaching notes and exercises are available on OUP web.

Skill *Problem-solving*

Sketching the waveform after a certain time

① Express the time interval in terms of period T.

② Find the distance (in terms of wavelength λ) that the wave travels in the time interval.

③ Displace the original waveform along the travelling direction by the distance obtained in ②. Then draw the remaining waveform.

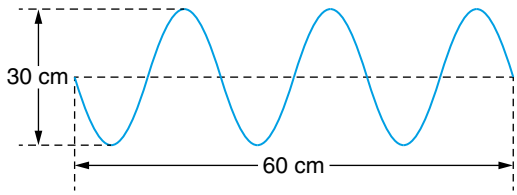
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Quick check 2

E-version in OUP Exercise Platform is also available.

1 Part of a travelling wave is shown below.



(a) Find its amplitude.

(b) Find its wavelength.

(c) It takes time t to produce 3 complete cycles of the wave. Complete the table.

	t / s	T / s	f / Hz
(i)	3		
(ii)	0.6		
(iii)		0.5	
(iv)			10

Learn by practice 1

1 Define the following terms describing wave motion and suggest a unit for each of them.

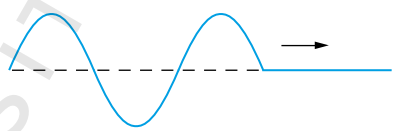
(a) Amplitude

(b) Period

(c) Frequency

(d) Wavelength

2 A wave with $T = 1$ s travels to the right. Its waveform at a certain instant is as shown. Sketch the waveform 0.5 s later.



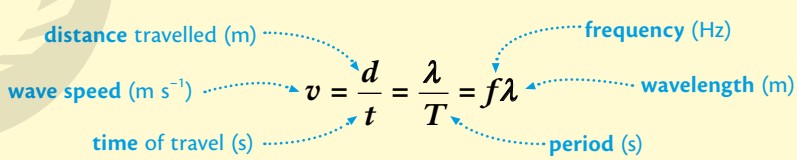
e Wave speed (v)

We may want to know how fast a wave travels. In this case, we need to find its **wave speed** v. Its unit is m s^{-1}.

The wave speed of a wave is the distance travelled by the wave in one second.

As discussed on p.8, a wave takes one period T to travel one wavelength λ.

By $v = \frac{d}{t}$ and $f = \frac{1}{T}$,



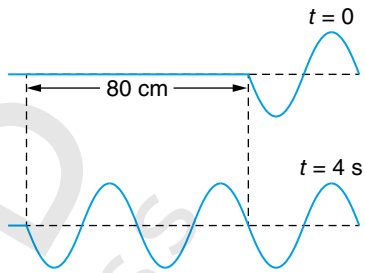
Ts may point that f depends on the vibrating source while v depends on the medium, so the relation $v = f\lambda$ tells how the property of the wave, the vibrating source and the medium are related.

The wave speed is **constant** along a **uniform medium**, such as a uniform spring. Unless otherwise specified, we assume that the medium is uniform and the wave speed is constant.

wave speed 波速率

Example 3 | Wave speed

The waveforms of a wave at time $t = 0$ and $t = 4$ s are as shown.



(a) Find the wave speed.

(b) Find the wavelength of the wave.

(c) Find the frequency of the wave.

Solution

Caution
Beware of the unit.
80 cm = 0.8 m

(a) $v = \frac{t}{d} = \frac{0.8}{4} = 0.2 \text{ m s}^{-1}$

(b) $\lambda = \frac{0.8}{2} = 0.4 \text{ m}$

(c) By $v = f\lambda$,

$f = \frac{v}{\lambda} = \frac{0.2}{0.4} = 0.5 \text{ Hz}$

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Quick check 3

E-version in OUP Exercise Platform is also available.

1 Complete the following table about a travelling wave.

	T / s	f / Hz	λ / m	v / m s^{-1}
(a)	2		3	
(b)			0.5	1
(c)		4		3
(d)		20	0.1	

Learn by practice 2

1 One end of a string is flicked continuously for 5 s and 2 cycles of a wave are produced in the process. The wavelength of the wave is 25 cm.

(a) What is the frequency of the wave?

(b) What is the speed of the wave?

2 A tsunami caused by an earthquake reaches a city in 1.5 hours. The water waves have a wavelength of 105 m and a frequency of 2 Hz.

(a) Find the speed of the waves.

(b) Estimate the distance between the earthquake's centre and the city.

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→ This is a Video Quiz about the experiment 'Particle speed and wave speed'.

DSE goal

Determine factors affecting speed of wave along stretched string or spring.

DSE exam

19(1A)Q18, 20(1A)Q16

2 Speed of a wave along a stretched spring

How can you increase the speed of a wave along a stretched string or spring? Increase the amplitude? Increase the frequency? Let us find out the answer in Experiment 4a.

Video

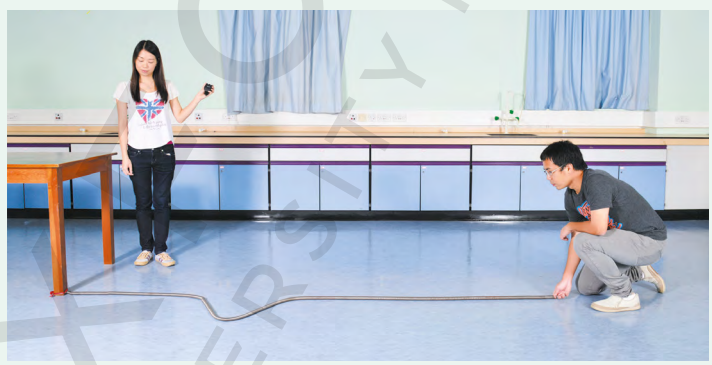
→ This video demonstrates Expt 4a.

→ This is a Video Quiz about this experiment.

Experiment 4a / Factors affecting the speed of a wave along a stretched spring

1 Fix one end of a long spring and stretch it on the floor. Measure the distance between the two ends.

2 Flick the free end to produce a transverse pulse. Measure the time needed for the pulse to travel to the other end. Calculate the pulse speed.



3 Repeat step 2 with pulses of different amplitudes.

4 Flick the spring repeatedly for a short time to produce a transverse wave. Increase the frequency of the wave by flicking the spring faster. Note how the wave speed changes with the frequency.

5 Repeat steps 1 and 2 by stretching the spring to different lengths.

Discussion

1 Does the wave speed change when (a) the amplitude, (b) the frequency of the wave change?

2 How does the wave speed change when the spring is stretched more?

✎ Stretching the spring more will give rise to a larger interacting force between the particles in the medium and a smaller mass will enable a shorter response time to disturbance, giving rise to a higher speed of propagation in both cases.

✎ There are some other media in which the wave speed is independent of the frequency. Such a medium is called a non-dispersive medium. For light wave, a vacuum is a non-dispersive medium.

The medium is the only factor affecting the wave speed.

▶ From the results of the experiment, the wave speed is **independent** of the **amplitude** and **frequency** of the wave. The wave speed increases when the spring is stretched longer. Further studies show that:

Wave speed = $\sqrt{\frac{T}{\mu}}$

▶ **A wave travels faster along a stretched spring (or string) when**
1 the tension T in the spring increases;
2 the mass per unit length μ of the spring decreases.

Example 4 Wave travelling along two strings connected together

A and B are two uniform strings of the same length, with B heavier than A. They are connected to form a long string. One end of the long string is attached to a pole. A student flicks the other end at a constant frequency to produce a wave as shown.



Explain how the wave speed, frequency and wavelength change as the wave travels from A to B.

Solution

If A and B have different tensions, a net force would act on the point of contact, making this point accelerate. ▶

A and B have the same tension as they are connected. B has a greater mass per unit length than A. Therefore, the wave speed v decreases when the wave travels from A to B.

The frequency f of the wave does not change because it is the same as the flicking frequency.

By $v = f\lambda$, λ decreases because v decreases and f remains constant.

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Quick check 4

E-version in OUP Exercise Platform is also available.

1 One end of a string is flicked to produce a transverse travelling wave. What will happen if the string is flicked with a higher frequency?

A The wave speed increases.

B The wave speed decreases.

C The wavelength increases.

D The wavelength decreases.

2 Which of the following changes can increase the wave speed along a spring? Put a tick in the corresponding box(es).

☐ Stretch the spring more to increase the tension in it.

☐ Replace the spring with a heavier one of the same length.

☐ Flick the spring with a larger amplitude.

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Exercise 4.2

1 A particle takes 2 minutes to make one complete vibration. What is the frequency of the vibration?

A 0.008 33 Hz

B 0.5 Hz

C 2 Hz

D 30 Hz

***2** A spring is flicked with frequency f to produce a transverse wave. Which of the following statements is/are correct if the spring is stretched to a longer length and flicked with the same frequency f?

(1) The period of the wave increases.

(2) The wave speed increases.

(3) The wavelength of the wave decreases.

A (2) only

B (3) only

C (2) and (3) only

D (1), (2) and (3)

***3** The waveform of a water wave with frequency 2.5 Hz is as shown (Fig a). Find the speed of the wave.

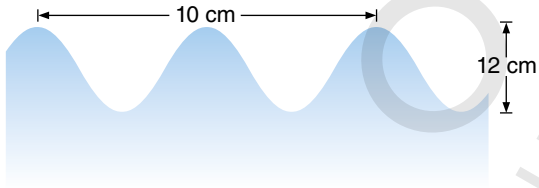


Fig a

A 4.8 cm s^{-1}

B 12.5 cm s^{-1}

C 25 cm s^{-1}

D 30 cm s^{-1}

4 Waves P, Q and R travel at the same speed. Their waveforms at a certain instant are shown (Fig b).

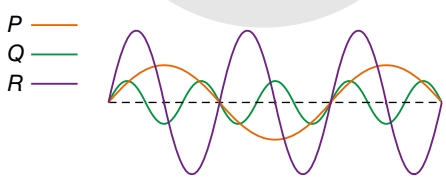


Fig b

Which of them has

(a) the largest amplitude,

(b) the longest wavelength,

(c) the highest frequency,

(d) the longest period?

***5** While a bird is drinking water, it produces 10 complete cycles of a wave in 5 s (Fig c). The wave travels 1 m in 5 s. Assume that the wavelength and the frequency of the wave remains unchanged.



Fig c

(a) Find the wavelength of the wave.

(b) Find the frequency of the wave.

(c) Find the speed of the wave.

***6** A wave completes 5 cycles in 2 s. Figure d shows its waveform at a certain instant.

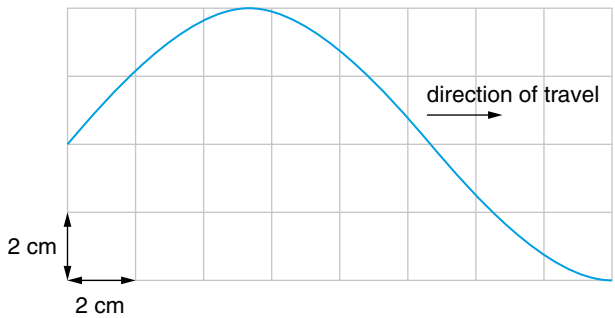


Fig d

(a) Find the amplitude of the wave.

(b) Find the wavelength of the wave.

(c) Find the period of the wave.

(d) Find the wave speed.

(e) Sketch the waveform after 0.1 s.

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4.3 Particle vibrations of transverse waves

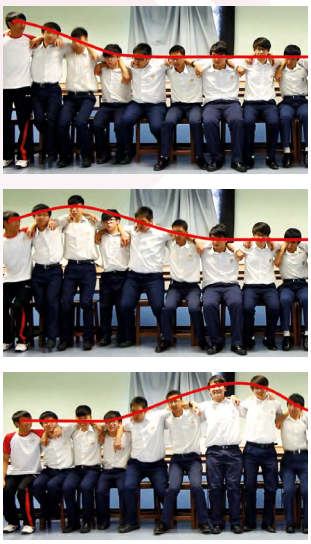


Fig 4.3a Students standing up and sitting down one by one.

Some students sit in a row. What will happen if they stand up and then sit down one by one (Fig 4.3a)?

A 'wave' can be seen travelling along the students. Similarly, when each particle in a medium continuously vibrates in a sequential manner, a wave will be formed. In this part, we will understand the relationship between wave motion and particle vibrations.

1 Wave motion and particle vibrations

Let us do the activity below to see how the particles on a wave move.

Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Transverse wave model

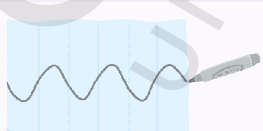
→ Worksheet is available on OUP web.

What you need



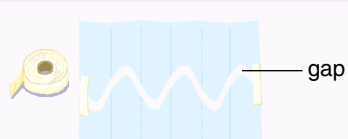
What to do

1 *X*



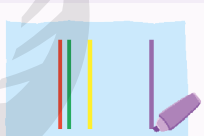
Draw a sine curve and vertical lines on paper *X*.

2



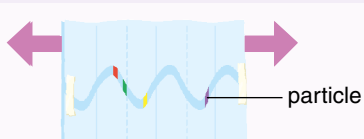
Cut the paper along the curve into 2 pieces. Connect the pieces back together with tape but leave a 3-mm gap in between.

3 *Y*



Draw several thick vertical lines in different colours on another paper *Y*.

4



Put *X* on top of *Y*. Move *X* sideways. The colour lines appear as 'particles' moving up and down as the 'wave' travels.

What you find

How are the vibrations of the 'particles' related to the motion of the 'wave'?

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In the above activity, the 'particles' vibrate up and down as the 'wave' moves (Fig 4.3b).

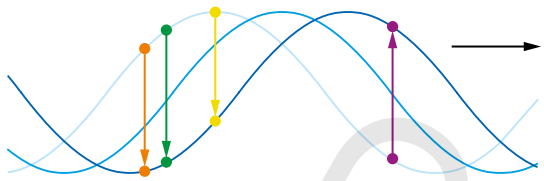


Fig 4.3b *Result of Hands-on activity.*

To study the wave motion and particle vibration more carefully, let us consider the waveforms of a wave and the particles on it at different times t (Fig 4.3c). The wave travels to the right and has a period T.

p and q each moves along a straight line. Their paths are drawn as loops for easy viewing.

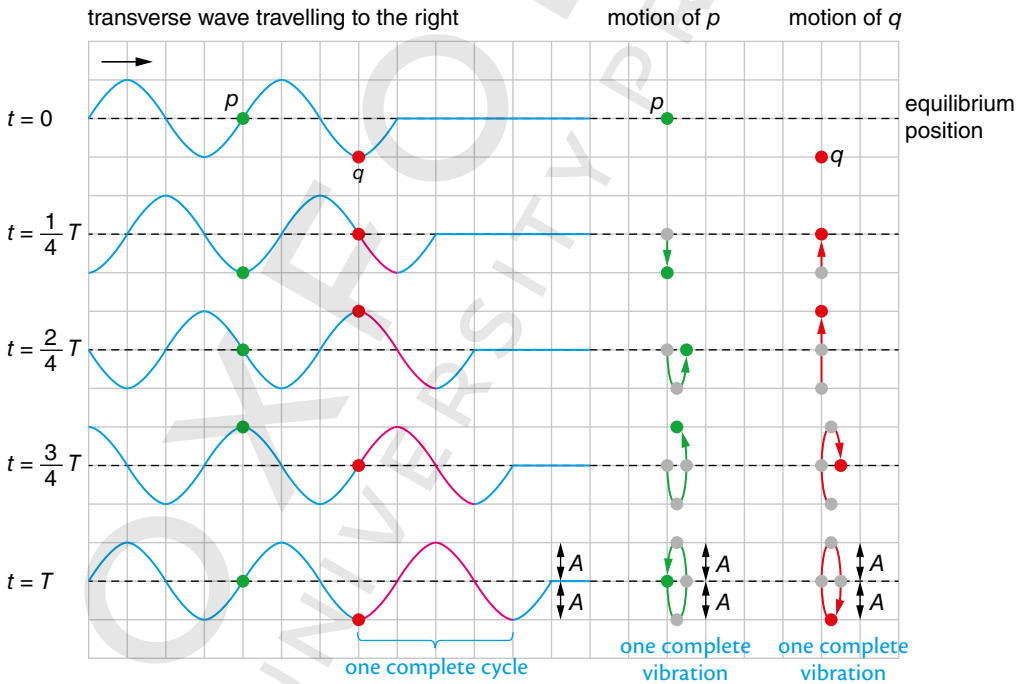


Fig 4.3c *Wave motion and particle motion.*

Simulation

→ This simulation shows the relationship between particle motion and wave motion in a transverse wave. Ss can observe the progression of the wave as one of its particles makes a complete vibration.

It can be observed that:

1 The size of the **maximum displacements** of p and q from their equilibrium positions, i.e. the amplitude, are equal to the amplitude of the wave.

Later on Ss may mix up amplitude and displacement and say 'maximum amplitude'.

2 The **time for p and q to complete one vibration**, i.e. the period, is equal to the period of the wave.

3 The **number of vibrations completed each second**, i.e. the frequency, by p and q are equal to the frequency of the wave.

On a transverse travelling wave, all particles and the wave have the same amplitude, period and frequency.

Quick check 5

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 A particle on a wave takes time t to move 10 cm from the highest position to the lowest position.

(a) What is the amplitude of the particle vibration?

(b) What is the distance travelled by the particle in one period?

(c) Complete the following table.

	t / s	T / s	f / Hz
(i)	1		
(ii)		0.5	
(iii)			10

2 A transverse travelling wave has an amplitude of 5 cm and a period of 4 s. Determine whether the following statements about the particles on the wave are true.

(a) Every particle has a displacement of zero in 4 s. (T/F)

(b) Every particle has a displacement of zero in 2 s. (T/F)

(c) Every particle has a frequency of 0.25 Hz. (T/F)

(d) A particle at a crest has a larger amplitude than a particle at the equilibrium position. (T/F)

DSE exam

14(1A)Q14, 16(1A)Q15

2 Motion of particles

Different particles on a wave may move in different directions at a certain instant. To determine their moving directions, we can consider the waveforms at a very short time before and after this instant.

Figure 4.3d shows a wave travelling to the right. At this instant:

- particles in green (a, b, h, i, j, p and q) are moving downwards;
- particles in red (d, e, f, l, m and n) are moving upwards;
- particles in yellow (c, g, k and o) are at the crests or troughs. They are reversing their moving direction and are momentarily at rest.

The situation is like throwing a ball into the air: at the instant when the ball is reversing its direction at the highest point, it is momentarily at rest.

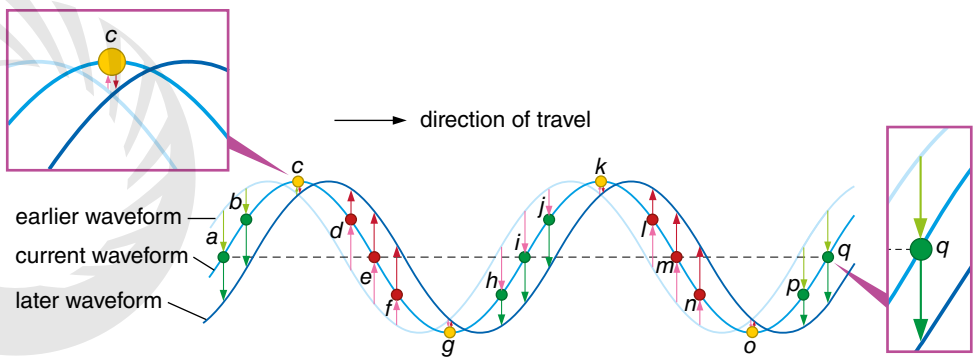


Fig 4.3d How the particles on a wave move.

Note that a particle keeps on moving at a varying speed—it reaches the **highest speed** at the **equilibrium position** and becomes **momentarily at rest** at a **crest** or a **trough**.

DSE goal
Describe wave motion in terms of phase.

3 Phase of particles

Figure 4.3e shows the paths of particles *W*, *X*, *Y* and *Z* on a transverse travelling wave in a certain time lapse.

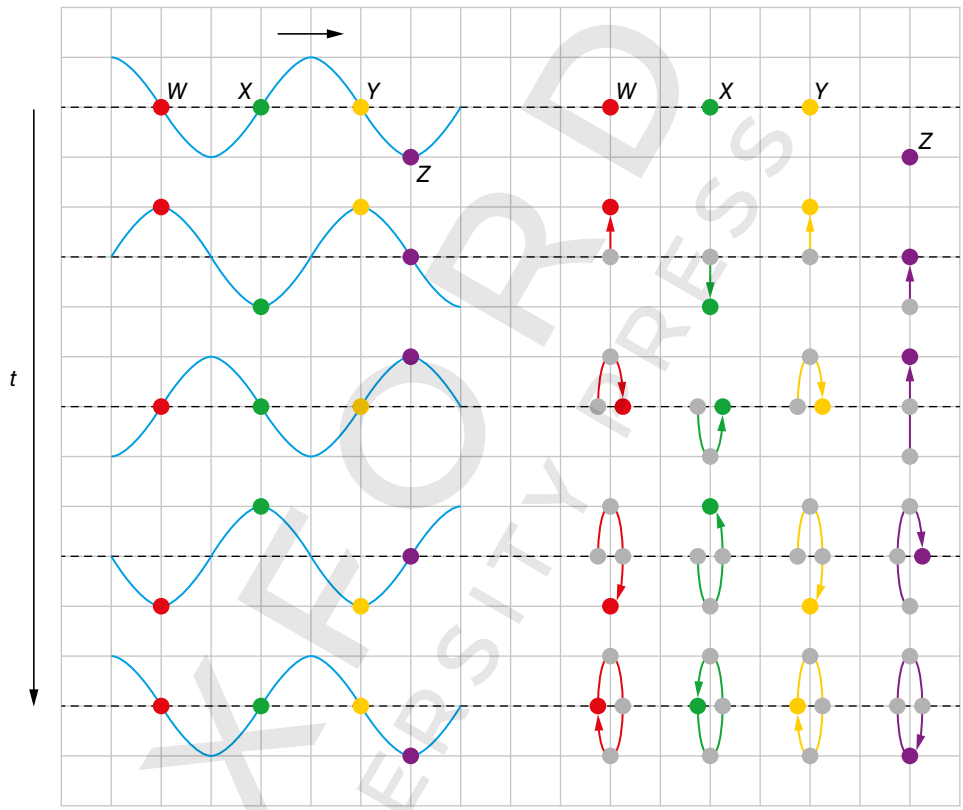


Fig 4.3e *Moving paths of particles.*

We can see the following relationship (Fig 4.3f):

✎ Remind Ss that the particles moving in the same direction do not necessarily vibrating in phase. Ss may overlook the criterion that the particles vibrating in phase must also have a separation of $n\lambda$.

1 Particles *W* and *Y* always move in the **same direction**. They vibrate **in phase**.

2 Particles *W*, *X* and *Z* do not vibrate in phase. They are said to be **out of phase**. In particular, *W* and *X* always move in **opposite directions**. They vibrate **in antiphase**.

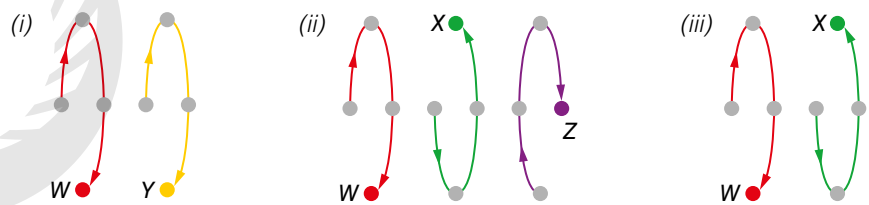


Fig 4.3f *Particles (i) in phase, (ii) out of phase, (iii) in antiphase.*

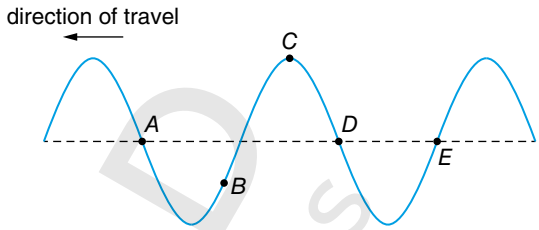
n is a positive integrer.

▶ In general, particles separated by 1λ, 2λ, …, $n\lambda$ are in phase. Particles with any other separations are out of phase. Particles separated by 0.5λ, 1.5λ, …, $(n - 0.5)\lambda$ are in antiphase.

in phase 同相　　out of phase 異相　　in antiphase 反相

Example 5 | Motion of particles on a wave

A transverse wave travels to the left along a string. *A*, *B*, *C*, *D* and *E* are particles on the string.



(a) Describe the motion of *A*, *B*, *C*, *D* and *E* at the instant shown.

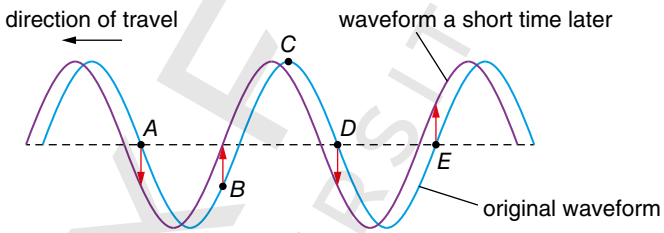
(b) Name two particles that are in phase.

(c) Name two particles that are in antiphase.

Solution

(a) *C* is momentarily at rest.

Consider the waveform a very short time later.



A and *D* are moving downwards. *B* and *E* are moving upwards.

(b) *A* and *D*

(c) *A* and *E* (or *D* and *E*)

Caution
By considering the waveform at a very short time later, we can determine the motion of any particle, except those at crests or troughs.

Tip
B and *E* are not in phase. Although they are moving upwards at this instant, they reach a crest at different instants.

Tip
A and *E* are 1.5λ apart.
D and *E* are 0.5λ apart.

Quick check 6

E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 A transverse wave travels to the right (Fig a).

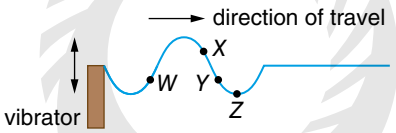


Fig a

True or false:

(a) *W* is moving upwards. (T/F)

(b) *W* and *Y* always reach their equilibrium positions at the same time. (T/F)

(c) *Z* is at rest forever. (T/F)

(d) *W* and *X* are in antiphase. (T/F)

(e) *W* moves with a constant speed. (T/F)

2 *P* is a particle on a travelling wave (Fig b).

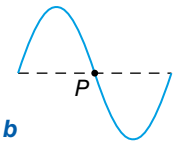


Fig b

If *P* is moving downwards at the instant shown, the wave is moving to the

A left.

B right.

Exercise 4.3

★**1** When a wave of amplitude x travels one wavelength λ, what is the displacement and the distance travelled by a particle on the wave?

	Displacement	Distance
A	0	$4x$
B	0	λ
C	$4x$	$4x$
D	λ	λ

★**2** Which of the following statements about the particles on a travelling wave is/are correct?

(1) Some particles have smaller amplitudes than the wave.

(2) A particle moves at the highest speed at a crest.

(3) Particles in antiphase move at the same speed.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

★**3** The positions of a particle on a transverse travelling wave at different instants are as shown (Fig a). The particle has a maximum displacement at position P.

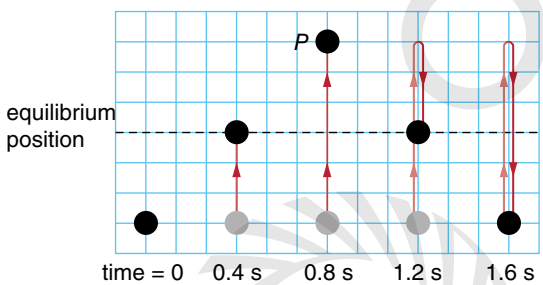


Fig a

Which of the following statements is/are correct?

(1) The particle has different amplitudes at different instants.

(2) The frequency of the wave is 0.8 Hz.

(3) The particle at P will return to P after 1.6 s.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

★**4** Figure b shows the waveform of a transverse wave at a certain instant. X, Y and Z are three particles on the wave. The wave travels to the right.

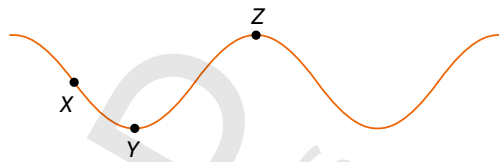


Fig b

Which of the following statements is/are correct?

(1) X is moving upwards at the instant shown.

(2) Y will travel to the existing position of Z after half a period.

(3) Z is moving downwards at the instant shown.

A (1) only

B (2) only

C (1) and (3) only

D (1), (2) and (3)

★**5** A wave travels to the right. P, Q, R, S and T are particles on the wave. The waveform at a certain instant is as shown (Fig c).

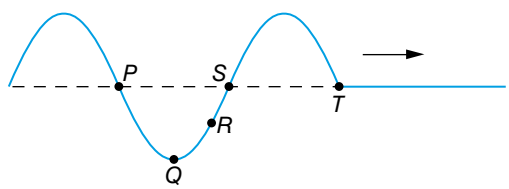


Fig c

(a) Sketch the waveform a very short time after this instant.

(b) Describe the motion of P, Q, R, S and T.

(c) Name one particle that vibrates in phase with P.

(d) Particle X vibrates in antiphase with Q. Indicate a possible position of X in the figure.

(e) Where is particle P after half a period?

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4.4 Graphical descriptions of transverse waves

In this part, we will learn about two kinds of graphs which can help us describe wave motion and particle vibration.

DSE goal

Present information on *s*–*d* graphs for travelling waves.

DSE exam

13(1A)Q16, 20(1A)Q12, 22(1A)Q15, 23(1A)Q14

This method can only be applied to transverse waves.

1 Displacement–distance graph of a wave

Consider a transverse wave produced on a spring. Figure 4.4a is a snapshot showing its waveform at a certain instant. If we copy the waveform and add grid lines and axes to it, we will obtain a **displacement–distance** graph (*s*–*d* graph) of the wave at this instant (Fig 4.4b).

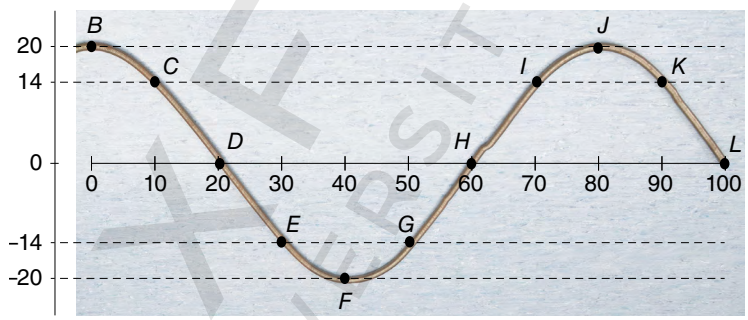


Fig 4.4a Snapshot of a wave on a spring.

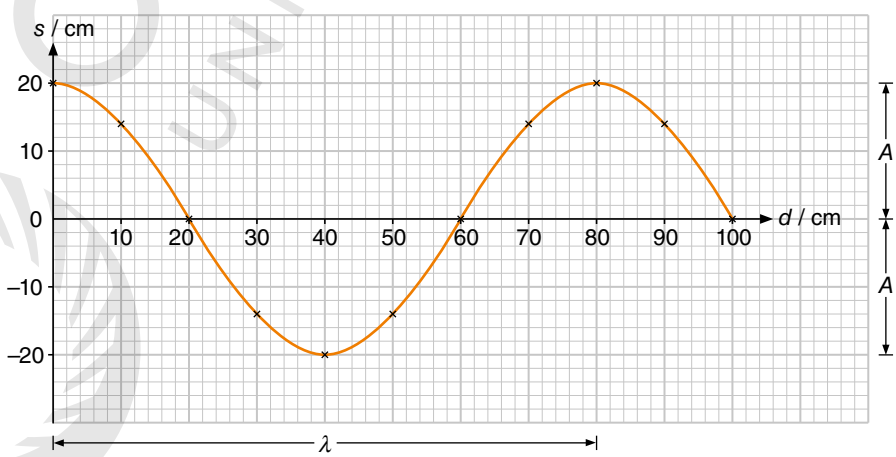


Fig 4.4b *s*–*d* of the wave.

In the graph, the *x*-axis shows the horizontal distances *d* of different particles on the wave from the reference point (particle *B*). The *y*-axis shows the displacements *s* of the particles from their equilibrium positions. Therefore, an *s*–*d* graph shows the displacements of **different particles** at a **certain instant**.

displacement–distance graph 位移—距離關係線圖

Note that:

1 The *s*–*d* graph of a transverse wave has the same shape as the waveform if we take the **leftmost** particle as the reference point.

2 We can read the **amplitude** A and **wavelength** λ from an *s*–*d* graph, but we cannot determine the period, frequency, wave speed or direction of travel of the wave just from the graph.

3 The *s*–*d* graph of a wave changes with time.

→ Teaching notes and exercises are available on OUP web.

Skill *Graph*

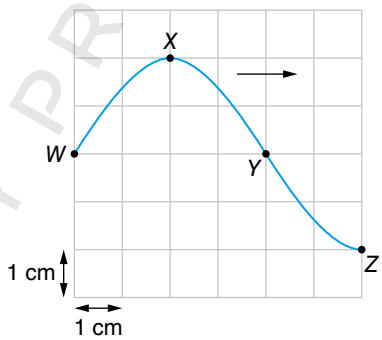
Constructing an *s*–*d* graph

Find out the following:

1 Amplitude
2 Wavelength
3 Location of equilibrium positions, crests and troughs (to determine the orientation of the curve)

Example 6 *s*–*d* graphs at different instants

The figure shows the waveform of a section of a wave at time $t = 0$. W, X, Y and Z are particles on the wave. The period of the wave is 4 s.

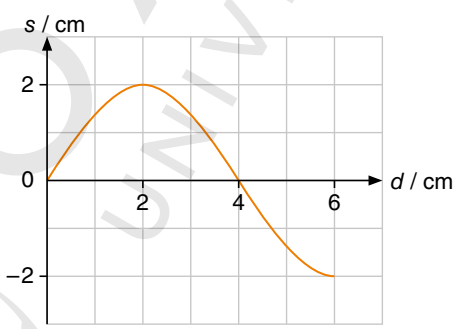


(a) Sketch the *s*–*d* graph of the wave from particles W to Z at
  (i) $t = 0$,
  (ii) $t = 1$ s.

(b) Find the speed of the wave.

Solution

(a) (i)



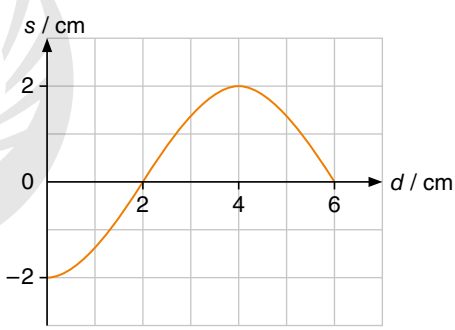
Tip

The leftmost particle W is used as the reference point, so the *s*–*d* graph has the same shape as the waveform.

$A = 2$ cm

$\lambda = 4 \times 2 = 8$ cm

(ii)



Tip

$T = 4$ s $\Rightarrow$ 1 s = 0.25T

The wave has travelled 0.25λ to the right.

$\therefore$ Y ($d = 4$ cm) is at a crest.

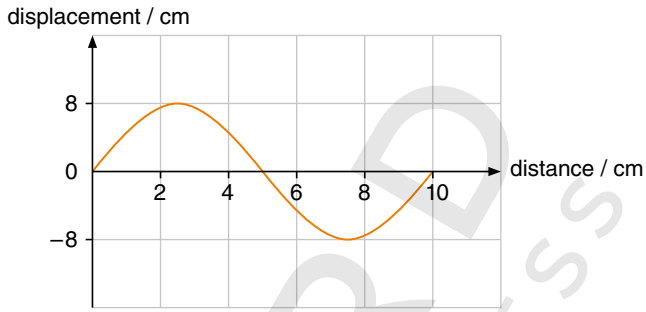
(b) $v = \dfrac{\lambda}{T} = \dfrac{0.08}{4} = 0.02 \text{ m s}^{-1}$

Google Forms Microsoft Forms

Quick check 7

E-version in OUP Exercise Platform is also available.

(For Q1–3.) The displacement–distance graph of a wave is as shown.



1 What is the amplitude of the wave?

A 5 cm B 8 cm

C 10 cm D 16 cm

2 What is the wavelength of the wave?

A 5 cm B 8 cm

C 10 cm D 16 cm

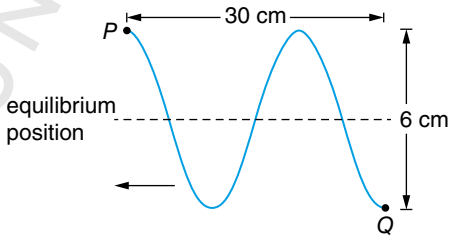
3 If the wave speed is 20 cm s^{-1}, what is the period of the wave?

A 0.02 s B 0.5 s

C 2 s D 5 s

Learn by practice 3

1 A wave travels to the left and its waveform at a certain instant is as shown. The period of the wave is T.



Sketch the displacement–distance graph from P to Q

(a) at this instant,

(b) after $\frac{1}{2}T$,

(c) after $\frac{3}{4}T$.

2 A wave has a wavelength of 10 cm and an amplitude of 4 cm. P and Q are two particles on the wave and their equilibrium positions are 15 cm apart. P is at a crest at time $t = 0$. Sketch the s–d graph of the wave from P to Q at $t = 0$.

DSE goal

Present information on *s–t* graphs for travelling waves.

DSE exam

15(1A)Q13, 18(1A)Q15, 21(1A)Q11, 23(1A)Q15

2 Displacement–time graph of a particle

A particle on a wave vibrates continuously. Its displacement s from its equilibrium position keeps changing with time t. By plotting s against t, we obtain the **displacement–time graph** (s–t graph) of that particle. It shows the displacements of **a particle** at **different instants**.

Figure 4.4c shows the s–t graph of a particle on a wave. This particle is at the equilibrium position and moving upwards at $t = t_0$. Note that:

1. We can read the **amplitude** A and **period** T from an s–t graph, but we cannot determine the wavelength, wave speed or direction of travel of the wave just from the graph.
2. The s–t graphs of different particles on the same wave have the same amplitude and period but different phases.

Caution

Never draw the travelling direction of the wave in an s–t graph. The particle does **not** travel in the same direction as the wave!

Emphasize that although s–d graph and s–t graph look very similar, they focus on different aspects of a wave.

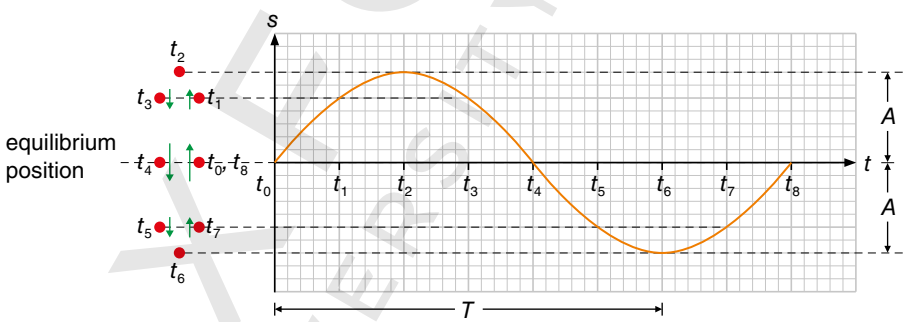
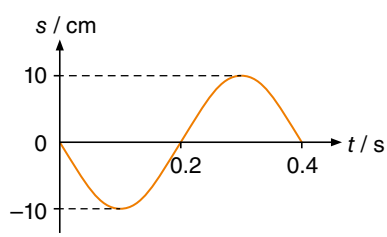


Fig 4.4c *The s–t graph of a particle on a wave (upwards as positive).*

Example 7 Reading an *s–t* graph

The s–t graph (upwards as positive) of a particle on a wave is as shown.



(a) What is the amplitude of the wave?

(b) To which direction is the particle moving at $t = 0.2$ s?

(c) If the wavelength of the wave is 50 cm, what is the wave speed v?

Solution

(a) 10 cm

(b) Positive direction (upwards)

(c) $v = \frac{\lambda}{T} = \frac{0.5}{0.4} = 1.25 \text{ m s}^{-1}$

Tip

s is increasing at $t = 0.2$ s.

displacement–time graph 位移—時間關係線圖

→ Teaching notes and exercises are available on OUP web.

Skill
Graph

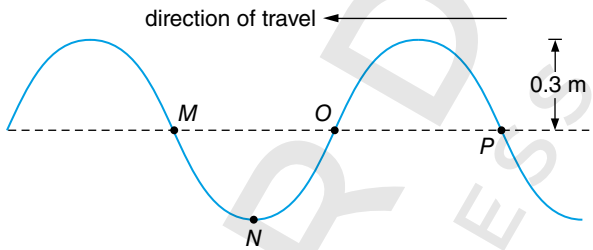
Constructing an *s*–*t* graph

Find out the following:

1 Amplitude
2 Period of vibration
3 Moving direction of the particle at a certain instant

Example 8 Drawing *s*–*t* graphs of different particles

A transverse wave travels at 0.5 m s^{-1} towards the left. The period of the wave is 0.2 s. Its waveform at time $t = 0$ is as shown. M, N, O and P are particles on the wave.



Take the upward direction to be positive.

(a) Sketch the *s*–*t* graph of particle M from $t = 0$ to $t = 0.4$ s.

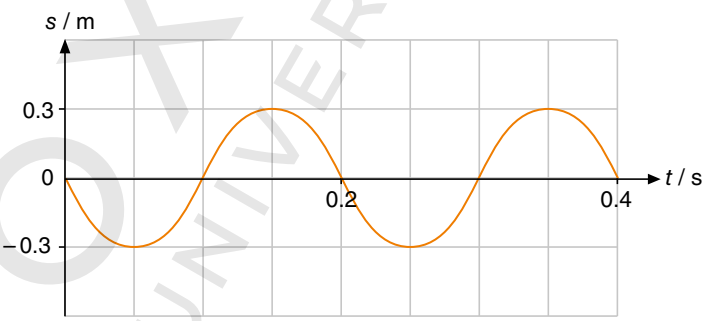
(b) Sketch the *s*–*t* graphs of particles O and P on the same graph obtained in (a).

(c) Find the wavelength λ of the wave.

Solution

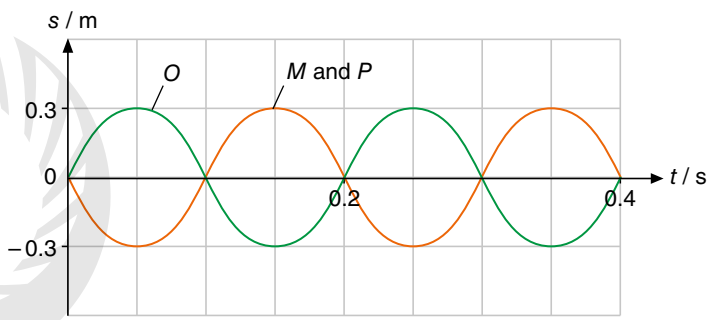
Tip
$A = 0.3$ m, $T = 0.2$ s
At $t = 0$, $s = 0$
Wave travels to the left
⇒ M moving downwards
⇒ s becomes negative

(a)



Tip
P in phase with M
⇒ same *s*–*t* graph
O in antiphase with M
⇒ moving upwards at $t = 0$

(b)



(c) $\lambda = vT = 0.5 \times 0.2 = 0.1$ m

Think about...

Does the *s*–*t* graph of N depend on the travelling direction of the wave?

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Although an *s*–*d* graph and an *s*–*t* graph look very similar, they provide different information about a wave (Table 4.4a).

	***s*–*d* graph**	***s*–*t* graph**
Amplitude A	✓	✓
Period T	✗	✓
Wavelength λ	✓	✗

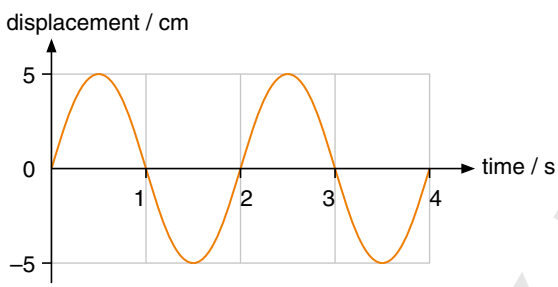
Table 4.4a *Comparison between an s–d graph and an s–t graph.*

Quick check 8

E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

(For Q1–4.) The displacement–time graph of a particle on a travelling wave is as shown.



1 What is the amplitude of the wave?

A 2 cm

B 5 cm

C 10 cm

2 What is the period of the wave?

A 1 s

B 2 s

C 4 s

3 What is the frequency of the wave?

A 0.25 Hz

B 0.5 Hz

C 1 Hz

4 If the wave speed is 5 cm s^{-1}, what is the wavelength?

A 2.5 cm

B 10 cm

C 20 cm

Learn by practice 4

1 A particle P on a transverse travelling wave vibrates with an amplitude of 4 cm and a period of 0.5 s. It is at a trough at time $t = 0$.

(a) Sketch the *s*–*t* graph for this particle from $t = 0$ to $t = 1$ s.

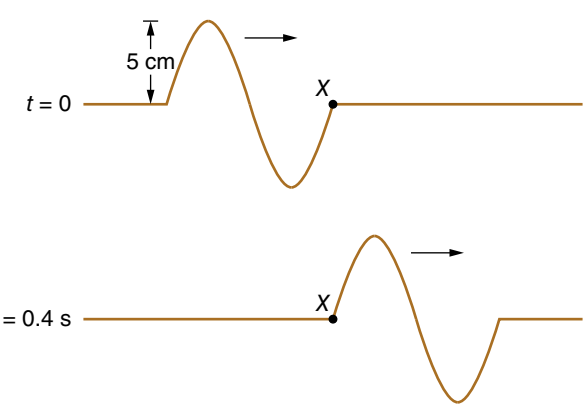
(b) Can you determine the wavelength from the *s*–*t* graph?

(c) Another particle Q on the wave vibrates in antiphase with P. Sketch the *s*–*t* graph of Q from $t = 0$ to $t = 1$ s.

2 A pulse travels along a stretched string. It reaches particle X on the string at time $t = 0$ and leaves X at $t = 0.4$ s as shown.

(a) Does X move upwards or downwards at $t = 0.01$ s?

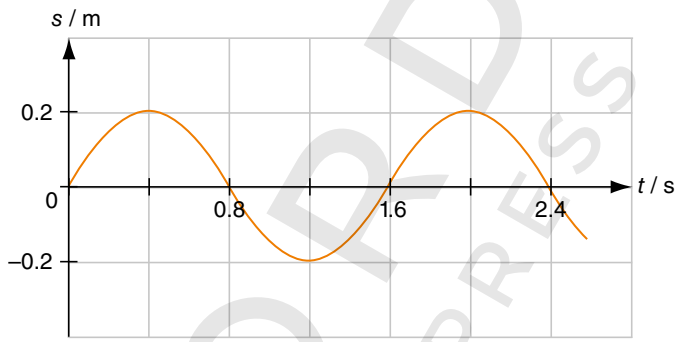
(b) Sketch the displacement–time graph of X from $t = 0$ to $t = 0.4$ s.



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Example 9 Different graphs of wave motion

A vibrator is connected to one end of a long string. It vibrates to generate a transverse travelling wave of wavelength 0.6 m along the string. The figure shows the s–t graph of particle P at a distance 0.9 m from the vibrator.



(a) Find the amplitude of the wave.

(b) Sketch the s–d graph for the wave from the vibrator to P at t = 0.8 s.

(c) Sketch how the kinetic energy of P changes from t = 0 to t = 1.6 s.

Solution

(a) 0.2 m

Tip

A = 0.2 m, λ = 0.6 m

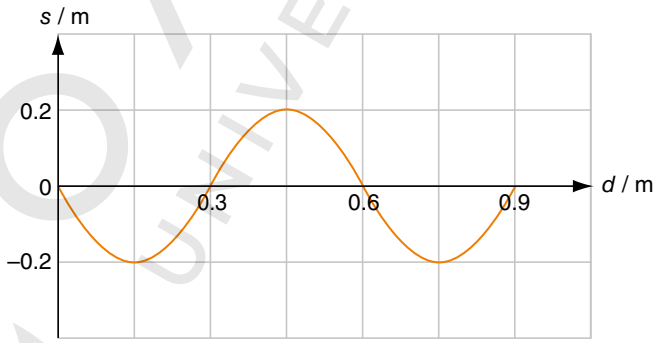
Consider P (d = 0.9 m).

At t = 0.8 s,

s = 0 and is decreasing

⇒ trough is approaching

(b)



Tip

s = 0 (at t = 0, 0.8 s, 1.6 s)

⇒ highest speed

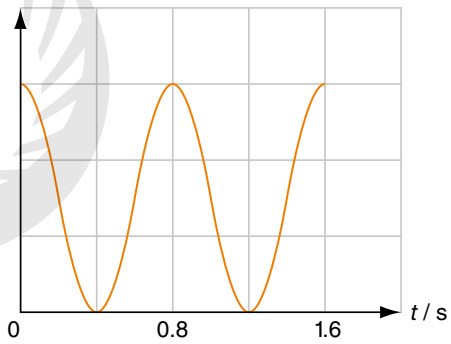
⇒ KE max

s = 0.2 m (at t = 0.4 s, 1.2 s)

⇒ momentarily at rest

⇒ KE = 0

(c)



Exercise 4.4 Q5 (p.29)

☆ HKDSE 2018 Paper 1A Q15

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | *s–t* graphs of two particles

One end of a long string is connected to a vibrator. P and Q are two particles on the string. The vibrator starts to vibrate at time $t = 0$ and generates a transverse travelling wave of wavelength 10 cm along the string. The s–t graph of P from $t = 0$ to $t = 10$ s is shown in Figure a. The s–t graph of Q from $t = 6$ s to $t = 10$ s is shown in Figure b. Upwards is taken as positive in the s–t graphs.

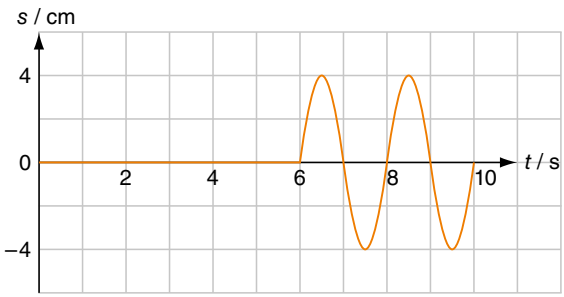


Fig a

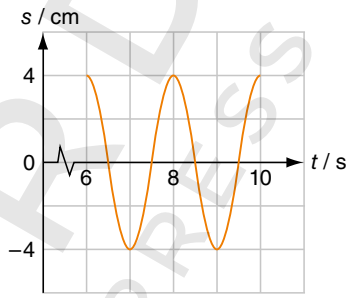


Fig b

Which of the following statements **must be** correct?

(1) P is moving upwards at $t = 8$ s.

(2) P is 30 cm from the vibrator.

(3) P and Q are 2.5 cm apart.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

Solution

The displacement of P is increasing at $t = 8$ s.

∴ P is moving in the positive direction, i.e. upwards, at $t = 8$ s.

∴ (1) is correct.

Common mistake
Students may not know how to determine the moving direction of a particle from its s–t graph.

$$v = \frac{\lambda}{T} = \frac{10}{2} = 5 \text{ cm s}^{-1}$$

As the wave takes 6 s to travel from the vibrator to P,

distance between P and vibrator $= vt = 5 \times 6 = 30$ cm

∴ (2) is correct.

Common mistake
Students may not notice that the time for the wave to travel from the vibrator to P can be read from the s–t graph.

At $t = 6$ s, P is at the equilibrium position and Q is at a crest.

Moreover, Q starts vibrating before P.

∴ Distance between P and $Q = \frac{1}{4}\lambda, \frac{5}{4}\lambda$ or $\frac{9}{4}\lambda$

Common mistake
Students may not be aware that there are several possible answers.

$d = 2.5$ cm is only one of the possible answers.

∴ (3) is not necessarily correct.

∴ The answer is A.

▶ Revision 4 Q11 (p.33), Q23 (p.36)

Exercise 4.4

★**1** P is a particle on a transverse travelling wave. Its s–t graph is shown below (Fig a).

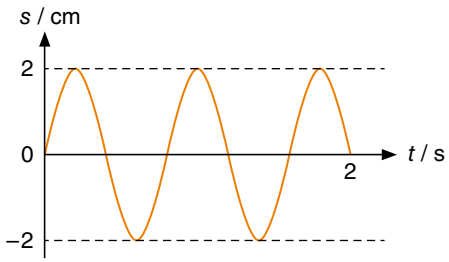


Fig a

Which of the following statements about the motion of P is/are correct?

(1) The amplitude of the vibration is 2 cm.

(2) The period of the vibration is 2 s.

(3) P changes its direction of motion every 0.8 s.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

★**2** The s–d graph of a travelling wave at time $t = 0$ is as shown (Fig b). W, X, Y and Z are particles on the wave. The wave travels from W to Z. The s–t graph of one of these particles is shown in Figure c.

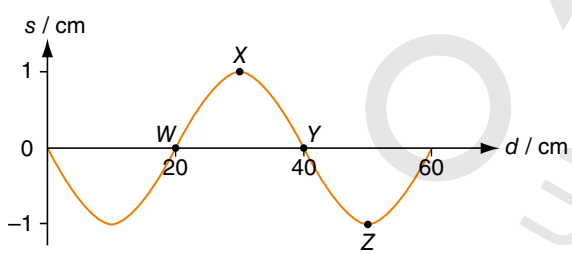


Fig b

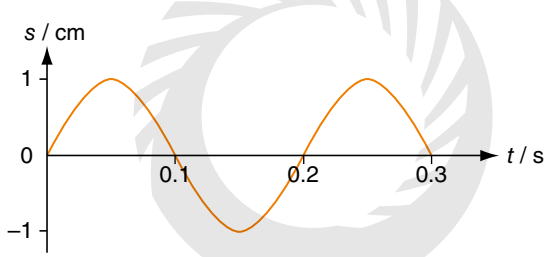


Fig c

(a) What is the wave speed?

(b) Figure c shows the s–t graph of which particle?

(c) Where is W at $t = 0.05$ s?

★**3** A vibrator O produces a wave on a string. The waveform at time $t = 0$ is as shown (Fig d). P and Q are two particles on the wave. The shortest time for P to reach the equilibrium position from the position shown is 0.1 s.

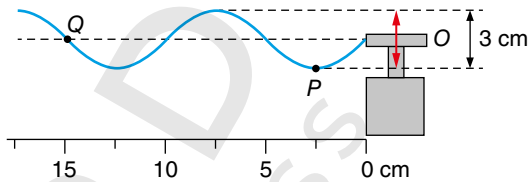


Fig d

Sketch the s–d graph of the wave from P to Q at $t = 0$ and $t = 0.3$ s. Indicate the positions of P and Q on the graphs.

★**4** Figure e shows two leaves X and Y floating on a calm water surface. The vibrator is switched on at time $t = 0$ and a wave is sent towards the leaves. The s–t graph of X is shown in Figure f.

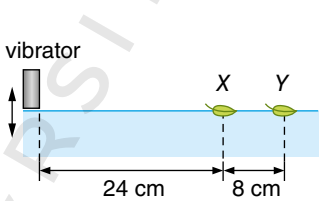


Fig e

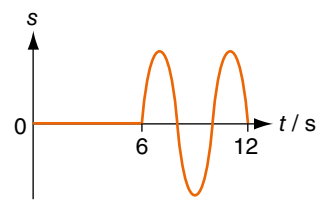


Fig f

(a) Find the speed of the water wave.

(b) Find the frequency and wavelength of the wave.

(c) Describe the motion of X at $t = 9$ s.

(d) Sketch the s–t graph of Y from $t = 0$ to $t = 12$ s.

(e) Sketch the waveform of the water wave from the vibrator to Y at $t = 8$ s.

★**5** A wave with wavelength 0.5 m is produced by a source. Particle Q is 0.75 m from the source. Figure g shows the s–t graph of Q.

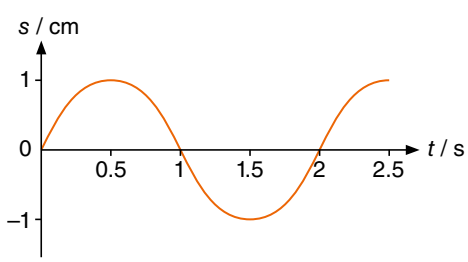
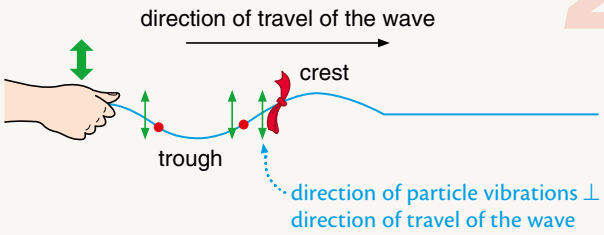


Fig g

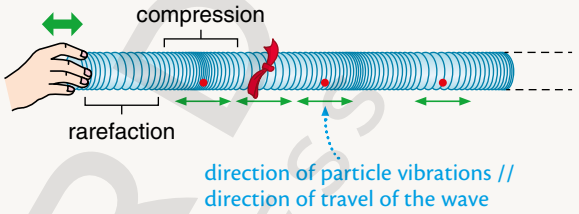
Sketch the s–d graph from the source to Q at $t = 1$ s.

Key concepts 4

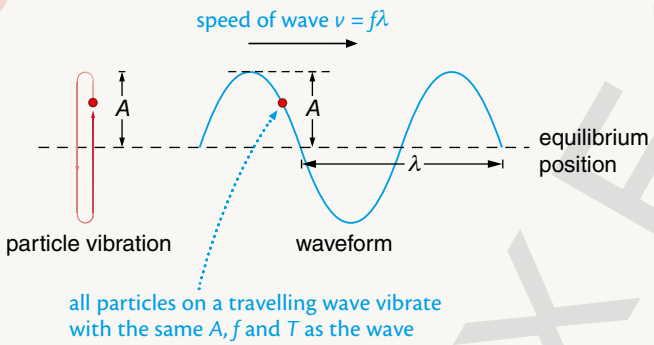
1 Transverse wave



2 Longitudinal wave

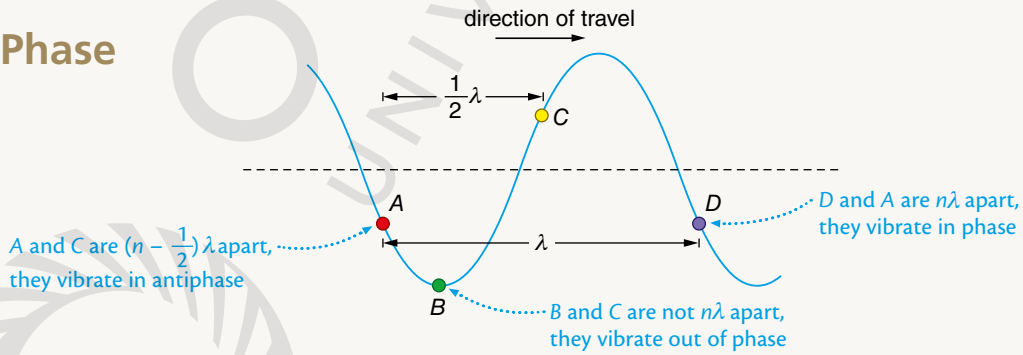


3 Describing wave

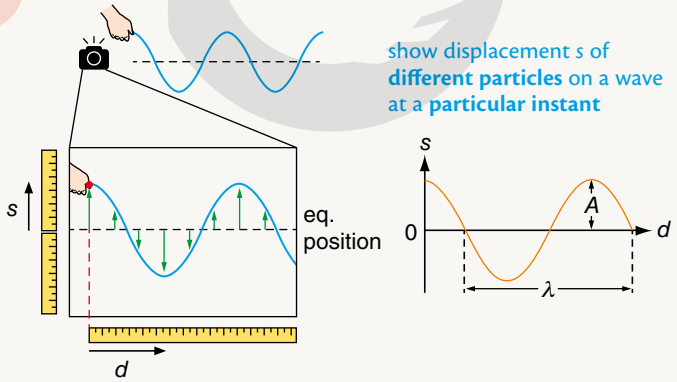


amplitude *A*	distance of crest or trough from equilibrium position
period *T*	time for one complete cycle of wave to be formed
frequency *f*	number of complete cycles of wave produced in 1 s
wavelength *λ*	minimum distance that the waveform repeats itself
wave speed *v*	distance travelled by wave in 1 s

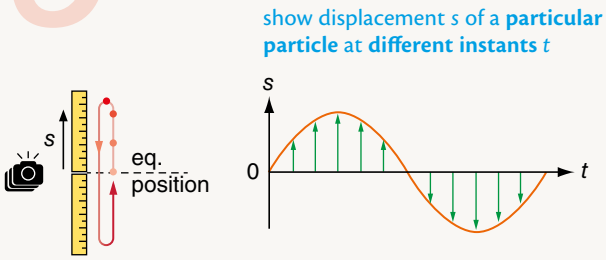
4 Phase



5 *s–d* graph



6 *s–t* graph



Key terms

1 amplitude 振幅 p.7
2 compression 密部 p.5
3 crest 波峯 p.4
4 displacement–distance graph 位移—距離關係線圖 p.21
5 displacement–time graph 位移—時間關係線圖 p.24
6 equilibrium position 平衡位置 p.7
7 frequency 頻率 p.8
8 hertz (Hz) 赫茲 p.8
9 in antiphase 反相 p.18
10 in phase 同相 p.18
11 longitudinal wave 縱波 p.4
12 out of phase 異相 p.18
13 period 週期 p.8
14 rarefaction 疏部 p.5
15 transverse wave 橫波 p.4
16 travelling wave 行波 p.2
17 trough 波谷 p.4
18 vibration/oscillation 振動 p.2
19 wave 波 p.2
20 wave speed 波速率 p.10
21 waveform 波形 p.4
22 wavelength 波長 p.7

Revision 4

Multiple-choice questions

1 4.2 Which of the following descriptions about the period of a transverse wave is **incorrect**?

A The time taken to produce one complete cycle of wave
B The time taken for a trough to travel one wavelength
C The time taken for a particle to travel four amplitudes
D The time taken for a particle to return to its equilibrium position

2 4.4 The *s*–*d* graph of a wave at a certain instant is shown in Figure a.

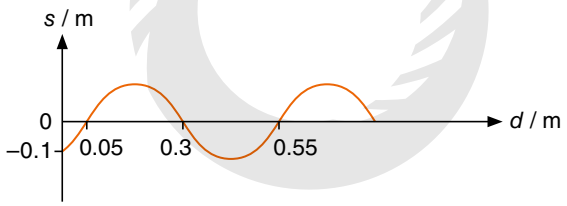


Fig a

If the wave travels at 20 m s^{-1}, what is its period?

A 0.0125 s
B 0.015 s
C 0.025 s
D 0.0275 s

★**3** 4.1 Which of the following statements about a travelling wave **must be** correct?

(1) The particles vibrate in the direction perpendicular to the direction of travel of the wave.
(2) It can be described as vibrations.
(3) No matter is transferred by the wave.

A (1) only
B (3) only
C (1) and (3) only
D (2) and (3) only

★**4** 4.3 A wave travels along a string to the right. The waveform at time $t = 0$ is shown in Figure b. *P* to *U* are particles on the string.

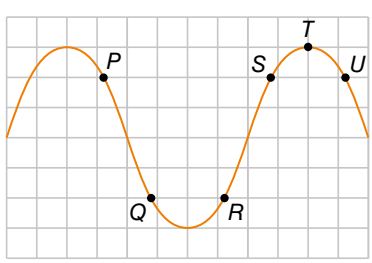


Fig b

Which particles are moving in the opposite direction as *P* at $t = 0$?

A *Q* and *U* only
B *R* and *S* only
C *R*, *S* and *T* only
D *Q*, *R* and *U* only

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★5 4.4 A transverse travelling wave has a frequency of 5 Hz. Its *s*–*d* graph at time $t = 0$ is shown below (Fig c). The distance d is measured from the vibrating source.

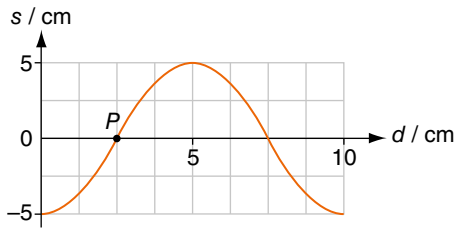


Fig c

Which of the graphs in Figure d is the correct *s*–*t* graph of particle *P* from $t = 0$ to $t = 0.2$ s?

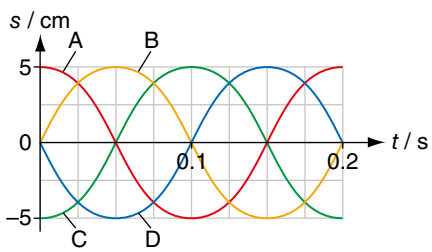


Fig d

★6 4.4 Figure e shows the *s*–*d* graph of a travelling wave at a certain instant. Figure f shows the *s*–*t* graph of a particle on the wave.

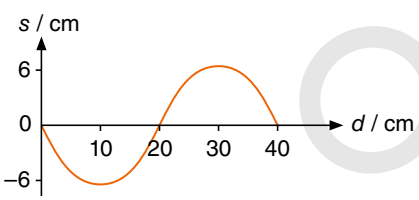


Fig e

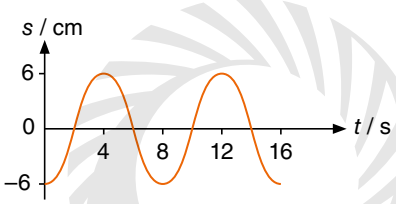


Fig f

How far does the wave travel in 4 s?

A 6 cm

B 10 cm

C 12 cm

D 20 cm

★7 4.3 The snapshots of a travelling wave at time $t = 0$ and $t = 0.6$ s are shown below (Fig g). The wave is travelling to the right.

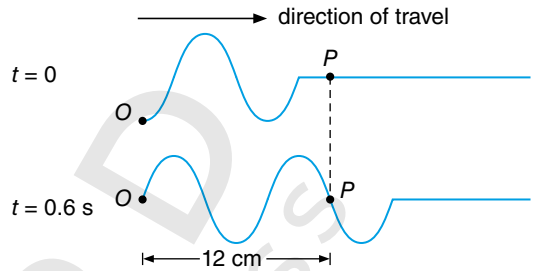


Fig g

Which of the following statements are correct?

(1) The frequency of the wave is 1.25 Hz.

(2) *P* starts to vibrate at $t = 0.2$ s.

(3) The wavelength of the wave is 6 cm.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★8 Misc Figure h shows the position of particle *X* on a wave travelling to the left at time $t = 2$ s.

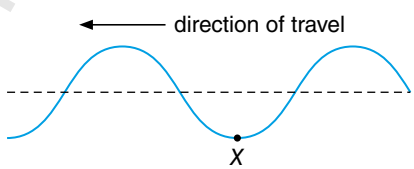
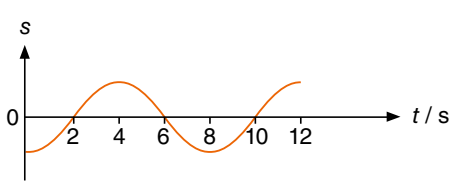


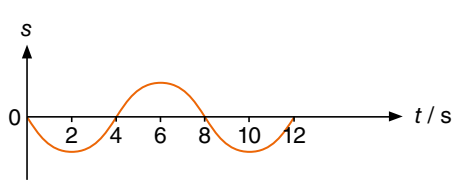
Fig h

Which of the following is a possible *s*–*t* graph of *X*? Take the upward direction as positive.

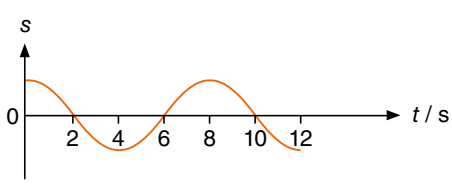
A



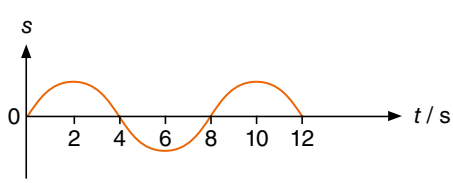
B



C



D



Q11 Exam report: Over half of the candidates chose options B and D which suggests they wrongly thought that the separation between the equilibrium positions of P and Q was 0.25λ.

***9** 4.3 A wave of frequency 3 Hz travels at 12 m s^{-1} through a medium. The distance between the equilibrium positions of particles P and Q on the wave is 5 m. What is the position of Q when P is on a crest?

A On a crest

B On a trough

C At its equilibrium position

D None of the above

****10** 4.2 A bird lands on a floating buoy. The buoy starts to produce a wave at time $t = 0$. Figure i shows a snapshot of the wave at $t = 0.6$ s.

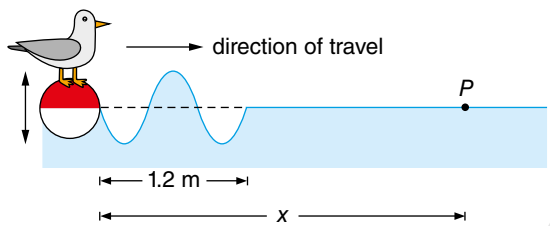


Fig i

Particle P first reaches its highest position at $t = 1.8$ s. What is the distance x between P and the buoy?

A 2.4 m B 3.0 m

C 3.6 m D 4.2 m

Refer p.11

11 4.4 **HKDSE 2018 Paper 1A Q15**

Figure j shows the displacement–time graph of particles P and Q on the same transverse travelling wave of wavelength λ.

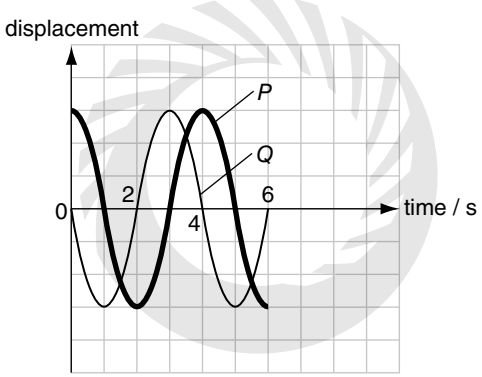


Fig j

Which of the following statements **must be** correct? Upward displacement is taken to be positive.

(1) At time $t = 2$ s, P is momentarily at rest.

(2) At time $t = 4$ s, Q is moving downwards.

(3) The separation between the equilibrium positions of P and Q is 0.25λ.

A (2) only

B (3) only

C (1) and (2) only (32%)

D (1) and (3) only

12 4.4 **HKDSE 2020 Paper 1A Q12**

Figure k shows part of the displacement–distance graph of a travelling wave of period T at time $t = 0$. P is a particle on the wave.

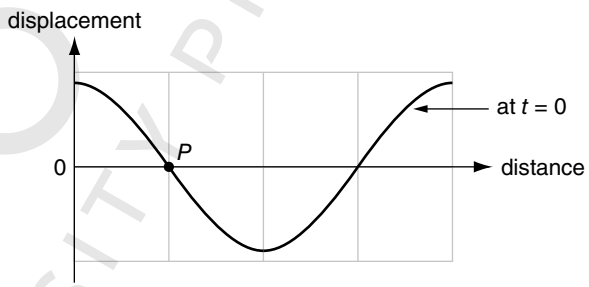
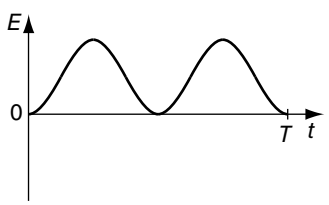


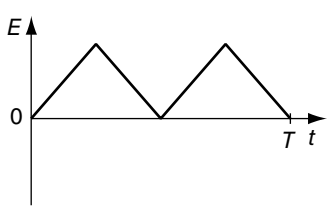
Fig k

Which graph below correctly shows the variation of the particle's kinetic energy E within a period starting from $t = 0$?

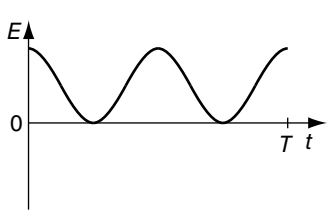
A



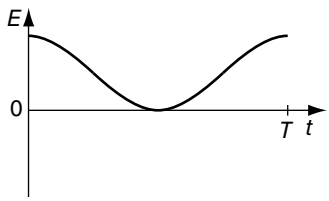
B



C (45%)



D



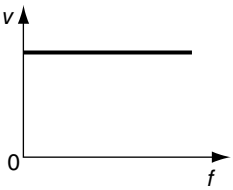
Exam report: Less than half of the candidates were able to obtain the correct answer of which the particle's kinetic energy is at a maximum at $t = 0$.

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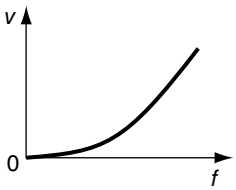
13 HKDSE 2020 Paper 1A Q16
4.4

A transverse wave propagates along a stretched string. Which graph below correctly shows the variation of the speed v of the wave with its frequency f?

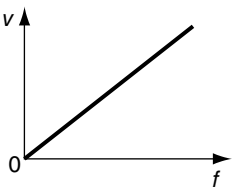
A (45%)



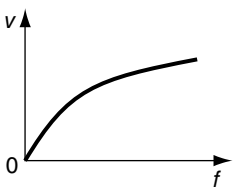
B



C



D



Conventional questions

14 ★
4.2

A student stretches a slinky spring to a length of 5 m. One end of the spring is fixed. She flicks the other end and observes that the transverse pulse takes 2 s to travel to the fixed end.

(a) Calculate the speed of the transverse pulse along the spring. (1 mark)

(b) If the student flicks the spring continuously at 4 Hz, what is the wavelength of the wave produced? (2 marks)

(c) Suggest **two** methods to produce a wave of shorter wavelength on the spring. Explain your answer. (4 marks)

15 ★
Misc

Figure l shows a water wave travelling to the right. The crest at X takes 2 s to reach Y. P and Q are two particles floating on the water. They are moving at the highest speed at the instant shown.

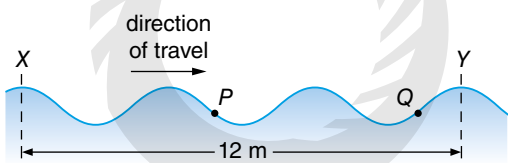


Fig l

(a) Find the wave speed. (2 marks)

(b) Find the frequency of the wave. (2 marks)

(c) Describe the motion of P and Q at the instant shown. (2 marks)

(d) How long will it take for P to first become momentarily stationary? (2 marks)

16 ★
Misc

Figures m and n show the s–d graphs of a transverse wave at time $t = 0$ and $t = 0.2$ s respectively. X, Y and Z are three particles on the wave.

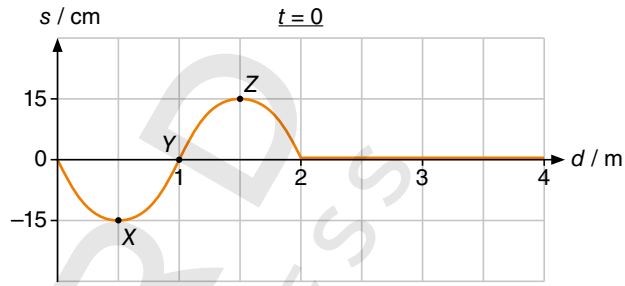


Fig m

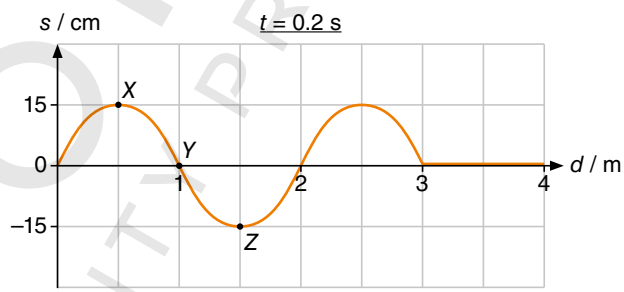


Fig n

(a) Find the amplitude, wavelength, wave speed and frequency of the wave. (4 marks)

(b) Which particle, X, Y or Z, is on a wave crest at $t = 0.6$ s? (1 mark)

(c) Describe the motion of X, Y and Z at $t = 0.5$ s. (3 marks)

17 ★
4.2

A transverse wave is produced in water and travels to the right. Figure o shows the waveform of the wave at time $t = 0$ and $t = 0.5$ s.

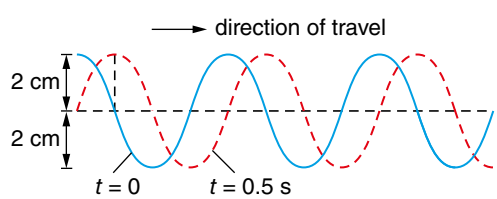


Fig o

(a) Find the amplitude of the wave. (1 mark)

(b) Explain what is meant by the wavelength of a wave. (2 marks)

(c) Suggest **two** possible frequencies for the wave. (4 marks)

Q13 Exam report: 40% of the candidates thought that the speed v increases linearly with frequency f and chose option C, which suggests that some candidates might have assumed the wavelength, instead of the speed, remains unchanged on a stretched string.

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*18 Misc Figure p shows a transverse travelling wave along a spring at time t = 0.5 s. Particle X on the wave completes 20 cycles of vibration in 4 s.

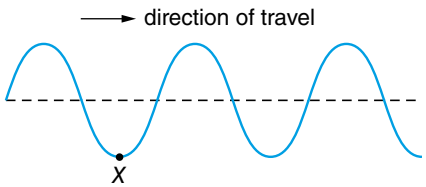


Fig p

(a) Particles X and Y on the wave vibrate in antiphase. Label all possible positions for particle Y in Figure p. (1 mark)

(b) Find the period of the wave. (2 marks)

(c) The minimum possible separation between the equilibrium positions of X and Y is 0.6 m. Find the speed of the wave. (2 marks)

(d) Sketch the s–t graph for X from t = 0 to t = 0.5 s. (2 marks)

(e) Sketch a graph to show how the kinetic energy of X changes from t = 0 to t = 0.5 s. (3 marks)

*19 4.2 You are provided with a long spring and a stop-watch (Fig q).

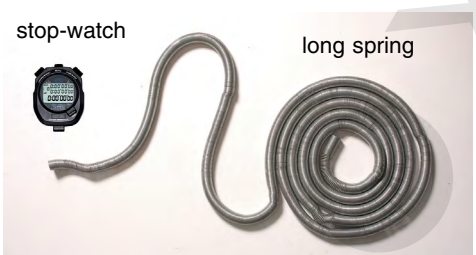


Fig q

(a) Describe how you can use the apparatus to demonstrate that wave speed is independent of the amplitude of a wave. (5 marks)

(b) Describe how you can increase the wave speed along the spring. (1 mark)

*20 Misc Figure r shows the waveform of a transverse wave at time t = 1 s. The wave travels to the right and has a period of 4 s.

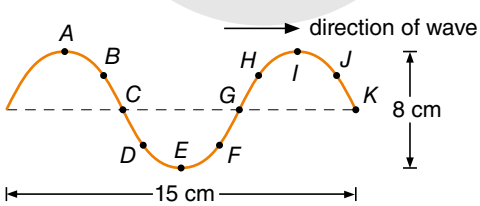


Fig r

(a) Which particles are moving in the same direction as particle C at t = 1 s? (4 marks)

(b) Sketch the displacement–time graph of particle C from t = 0 to t = 5 s. (3 marks)

**21 Misc A string is connected to a vibrator (Fig s). At t = 0, the vibrator starts to vibrate (upwards first). It produces a wave of amplitude 2 cm and period 0.2 s. P is a particle on the string 0.5 m away from the vibrator. The speed of the wave is 1 m s^{-1}.



Fig s

(a) In Figure s, sketch the waveform on the string at t = 0.3 s. (2 marks)

(b) Sketch the s–t graph of P from t = 0 to t = 1 s. (3 marks)

(c) What will happen to the wavelength and wave speed if the frequency of the vibrator is doubled? (2 marks)

Refer p.28

**22 Misc A wave of period 0.4 s travels along a string. Figure t shows the waveform of the wave at t = 0. Particle B is moving downwards at this instant.

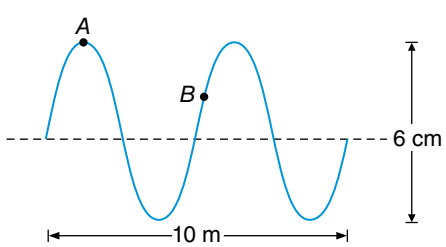


Fig t

(a) Find the amplitude and the wavelength of the wave. (2 marks)

(b) Find the speed of the wave. (2 marks)

(c) In Figure t, mark all positions at which particles are momentarily at rest at t = 0.1 s. (1 mark)

(d) Sketch the s–t graph of particle A from t = 0 to t = 0.5 s. (3 marks)

Refer p.25

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23 Misc **HKCEE 2009 Paper 1 Q10**

In a ripple tank, initially five tiny plastic beads (*P*, *Q*, *R*, *S*, *T*) float on the calm water surface. A vibrator begins to produce straight waves at time $t = 0$. Figure u shows the positions of beads on the waves at $t = 7$ s. Figure v shows the displacement–time graph of *S*.

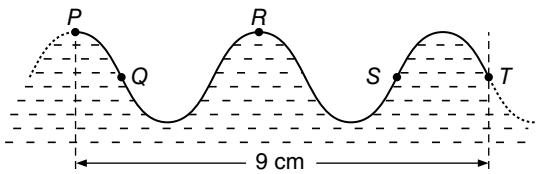


Fig u

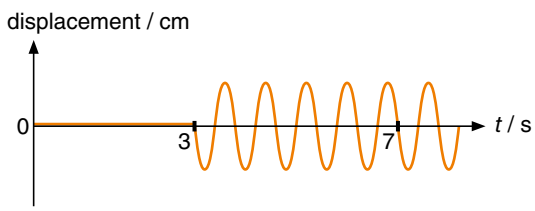


Fig v

(a) (i) Find the wavelength of the wave. (1 mark)

(ii) Find the frequency of the wave. (2 marks)

(iii) Find the distance between the vibrator and *S*. (3 marks)

(b) State the bead(s) that is/are moving

(i) in the same direction with *T* at time $t = 7$ s, (1 mark)

(ii) in the opposite direction with *T* at time $t = 7$ s. (1 mark)

(c) In Figure u, sketch the waveform between *P* and *T* at a quarter of period after $t = 7$ s. Mark the position of *S*. (2 marks)

Q23 Exam Report: This question was on wave motion in a ripple tank and candidates' performance was satisfactory.

(a) (i) Most candidates could work out the wavelength of the wave from the information given in the figure.

(ii) Some weaker candidates had difficulties in working out correctly the period of the wave and arrived at a wrong answer for the frequency *f*.

(iii) Many candidates did not know how to find the distance between the vibrator and *S*. The displacement–time graph was difficult to some weaker candidates and they did not know that 3 s was required for the water waves travelling from the vibrator to reach *S*.

(b) Most candidates were able to state correctly the beads moving (i) in the same direction and (ii) in the opposite direction with bead *T* at time $t = 7$ s.

(c) Many candidates could not draw correctly the waveform after a quarter of period. Some candidates did not sketch the waveform in Figure v as required.

24 Misc **CCEA GCSE Higher Tier 2014 Q1(c)**

Seismic (earthquake) waves travel through the earth to the surface. When they arrive there, they cause buildings on the surface to vibrate. One type of seismic wave, called an S-wave, causes buildings on the surface to vibrate parallel to the Earth's surface (Fig w).

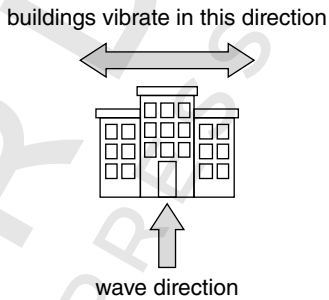


Fig w

(a) Seismic waves are either longitudinal or transverse. What type of wave is a seismic S-wave? (1 mark)

(b) Explain in reason for your answer to part (a). (2 marks)

(c) An S-wave has a speed of 3.5 km s^{-1} in the Earth's crust and its frequency is 1.3 Hz. Calculate its wavelength. Give your answer in metres. (4 marks)

(d) The vibrations caused by earthquakes are detected by a seismometer. The graph produced is called a seismograph. The diagram below shows a seismograph (Fig x).

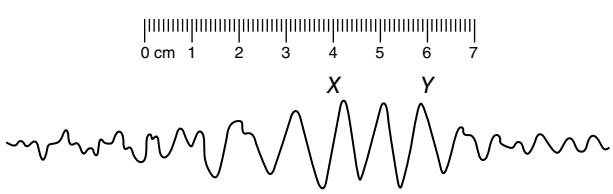


Fig x

Using the ruler shown in the diagram measure the horizontal distance between the peaks *X* and *Y*. Each 1 cm on the scale represents a time of 0.8 s.

Convert your measured distance to a time and use your answer to calculate the frequency of the seismic wave in the region *XY*. Remember the region *XY* shows two complete waves. (4 marks)

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